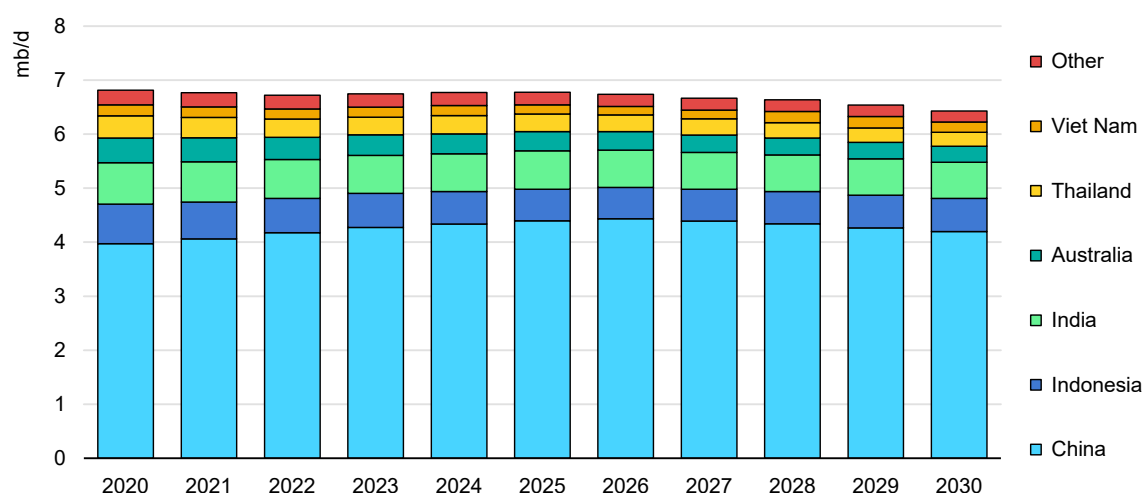


investments increasingly geared towards natural gas, signs of change are emerging with significant reforms under way in Indonesia and India.

China has been the exception, thanks to high reinvestment rates, exploration successes and a strong government mandate to increase output. The three state-owned Chinese oil companies – Sinopec, CNPC and CNOOC – have increased investments to stymie declines, raising production by 500 kb/d from the 2018 low of 3.8 mb/d to 4.3 mb/d in 2024. Recent, successive Chinese government Five-Year Plans¹ have laid out ambitious energy and climate goals that prioritise energy security and fossil fuel developments. The 15th Five-Year Plan, covering 2026-2030 looks to sustain annual oil production above 4 mb/d. Indeed, this Report sees continued offshore projects and onshore infill and delineation drilling efforts boosting total oil supply to a three-year plateau at 4.4 mb/d, before ending the decade at 4.2 mb/d.

Non-OPEC+ Asia oil supply slips through the decade, 2020-2030.



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Notes: Chart includes total oil production (crude, condensates, NGLs and non-conventional) from OECD and non-OECD Asia for countries that are not members of OPEC+. Other includes Bangladesh, East Timor, Mongolia, Myanmar, Nepal, New Zealand, Japan, Korea, Israel¹, Pakistan, Papua New Guinea and the Philippines.

Australian supply is forecast to fall from 370 kb/d to 300 kb/d in 2030, driven by declines in the Carnarvon Basin. Santos' 80 kb/d Dorado development was the most promising source of new production for Australia, yet the company recently deferred the project until after further evaluation of Bedout Basin resources has been completed.

¹ The statistical data for Israel are supplied by and under the responsibility of the relevant Israeli authorities. The use of such data by the OECD is without prejudice to the status of the Golan Heights, East Jerusalem and Israeli settlements in the West Bank under the terms of international law.

Indonesian oil production, currently at 590 kb/d, is expected to rise modestly to 620 kb/d by the end of the decade. This comes after a series of fiscal reforms, including increased producer revenue sharing to 75-95%, compared to the previous scheme where operators could, at times, receive zero revenue. These reforms, as well as expedited timelines for exploration wells and enhanced oil recovery (EOR) optimisation along with new commitments by Petronas, Eni and PetroChina all support production growth with upside potential over the medium term to this *Report's* forecast.

India also reformed its fiscal terms and held its tenth round of the Open Acreage Licensing Policy (OALP X). The new fiscal terms seek to modernise the upstream regulatory framework, ensuring greater stability and predictability. The windfall tax was removed, and early production fiscal incentives were introduced, while the act was broadened to include light tight oil and shale gas under the regulatory framework.

New tie-ups between Indian state-owned firms and western companies have provided tailwinds for the recent optimism felt in the sector. Examples include Oil India Ltd signing deals with both Petrobras and TotalEnergies, and state-owned Oil and Natural Gas Corporation Ltd (ONGC) agreeing to use a BP subsidiary for technical services. Indian production is expected decline marginally by 30 kb/d to 670 kb/d by the end of the decade.

Qatari oil supply gained 40 kb/d last year to reach 1.8 mb/d. Output is set to rise by 610 kb/d from 2024 to 2030, largely from associated condensates produced from Qatargas's four new LNG trains at North Field East (NFE) and two new trains at the North Field South (NFS) LNG expansion. The six new LNG trains are anticipated to bring on a total of 370 kb/d of condensate and additional associated NGLs online, starting with the first two NFE trains in mid-2026. Qatar's crude production is expected to edge up to 650 kb/d by 2030, from around 620 kb/d currently, largely due to works to lift output from the Al Shaheen field by North Oil Company, a joint venture between TotalEnergies and QatarEnergy.

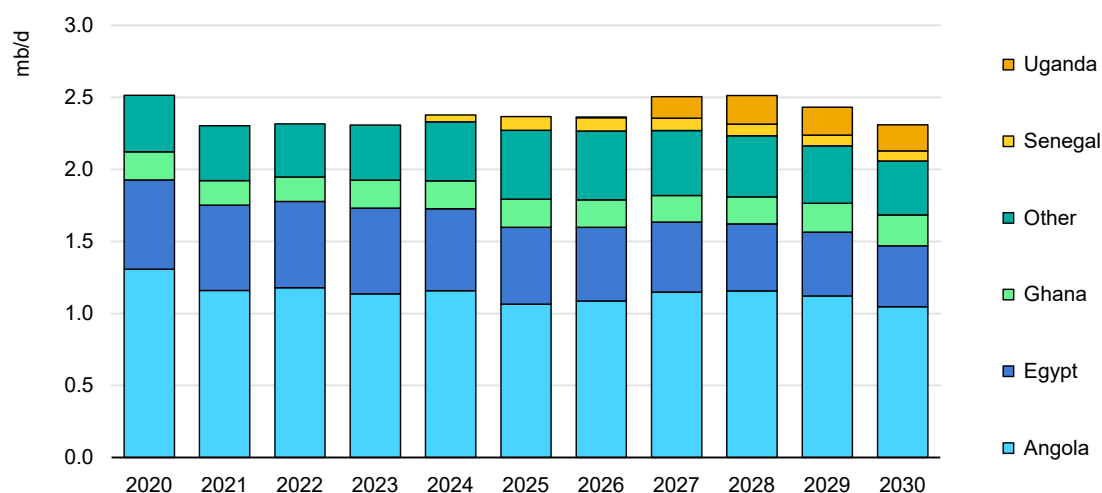
Cautious development in Africa

Non-OPEC+ African supply remains at the crossroads of resource availability and geopolitical risk. In Niger, the recently built pipeline to export additional production has been subject to terrorist attacks and cross-border arrests of high-level officials, resulting in low operating efficiency. Namibia's ruling party has proposed sectoral changes but there is ambiguity regarding their retroactive application to existing contracts. Meanwhile, Angola has pushed through contractual reforms to increase investment, expected to boost supply over the decade. Senegal and Côte d'Ivoire offer stable regimes as companies debate sanctioning more projects in their waters.

Overall, non-OPEC+ African production is forecast to remain relatively flat, falling by 70 kb/d, from 2.4 mb/d in 2024 to 2.3 mb/d by 2030. Despite over 600 kb/d of new capacity coming online, primarily split between Angola and Uganda, field declines continue to outweigh producers' best efforts at growing output. Additional supplies could come from the 580 kb/d of capacity that has not yet been sanctioned, providing some upside potential.

Angolan oil production has been declining for the better part of a decade as the country wrestles with mature fields and ageing infrastructure. After leaving OPEC in 2024, revamped fiscal terms have attracted renewed investments by different majors. Indeed, from 2019-24, 180 kb/d of new capacity was added in the country while 250 kb/d of sanctioned new capacity is forecast to come online through 2030 and another 100 kb/d is awaiting FID. Yet as one of Africa's largest offshore producers, these commitments are just enough to offset declines at existing fields through the decade. Production is set to increase by 100 kb/d through 2028, before returning to the current output of 1.1 mb/d in 2030.

Non-OPEC+ Africa total oil production is stable over the medium term, 2020-2030



IEA. CC BY 4.0.

Notes: Chart includes total oil production (crude, condensates, NGLs and non-conventional) from countries that are not members of OPEC+. Other includes Cameroon, Chad, Côte d'Ivoire, Democratic Republic of Congo, Mauritania, Mozambique, Niger, South Africa and Tunisia.

Namibia's Orange Basin has succumbed to geologic realities over the last year as some recent exploration wells have come up dry. Additional uncertainty entered the country's nascent oil and gas sector after recently elected president Netumbo Nandi-Ndaitwah placed the industry directly under her purview, ostensibly to streamline and speed up the permitting and development process.

The ruling party has also proposed increasing state carried interest in all developments. Two promising discoveries, the TotalEnergies-led Venus and the Galp-led Mopane, are in different phases of development, with TotalEnergies

slated to take FID next year while Galp is still looking for partners and working on a development plan. If both projects are developed, combined they could potentially produce around 250-400 kb/d sometime in the mid-2030s.

Elsewhere in West Africa, **Senegalese** volumes decline as Woodside is waiting to take FID on Sangomar Phase 2 following last year's 100 kb/d Sangomar FPSO start-up. Likewise, **Niger** supply declines from current levels without sustained infill drilling and delineation work in the Agadem Rift Basin. **Ghana**, on the other hand, sees a continued 20 kb/d uptick in production as drilling programmes there support offshore fields. Eni's success in **Côte d'Ivoire** has revitalised the country's oil and gas sector, leading to expected record in output in 2025. The company is looking to take FID on Baleine Phase 3 this year or next, which would add 100 kb/d, yet it is not included in our forecast. Additional, longer-term tailwinds come from their potential billion-barrel Calao discovery, still in the appraisal phase.

Uganda is set to see first oil in late 2026 from the two Lake Albert projects that TotalEnergies and CNOOC approved in 2021 and 2022, respectively. The developments have been marred by financing issues and environmental controversy as the crude will travel via a 1 440 km heated pipeline to the coast for export. Together, the Tilenga and Kingfisher fields have 200 kb/d of capacity.

Refining and trade

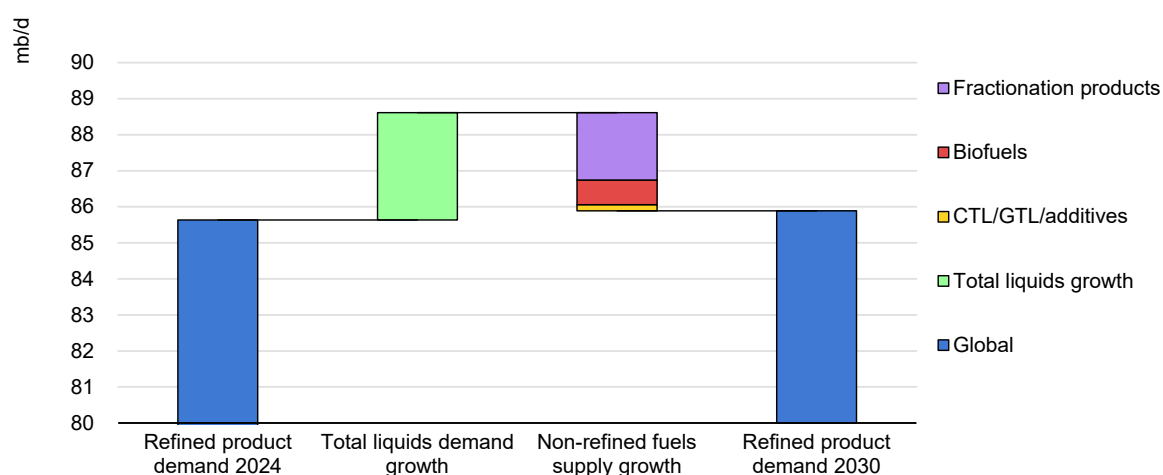
Global summary

Mounting challenges ahead for refining

The refining industry will be increasingly challenged by global oil demand growth underpinned almost exclusively by petrochemical feedstocks produced from non-refined products such as natural gas liquids (NGLs). Limited demand growth for refined products will be led by jet fuel and naphtha while gasoline and diesel post outright declines. This contrasts sharply with continued strong increases in biofuel supplies and NGLs, marginalising the role of refineries in meeting petrochemical feedstock demand.

For refiners, this implies a stark divergence from the long-term trend of rising transport fuel demand that has underpinned the industry's investment strategy and profitability for much of the past 50 years. Adapting to these shifting supply and demand dynamics presents several problems for the industry, but many refiners are already deftly navigating the evolving changes. These emerging trends are brought clearly into focus by our 2024-30 forecast.

Refined products demand growth, 2024-2030



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Notes: CTL = coal-to-liquids and GTL = gas-to-liquids. Refined product demand is net of CTL/GTL, NGLs, additives, biofuels and direct use of crude.

Refined products demand is projected to peak in 2027 at 86.3 mb/d, only 710 kb/d above 2024 levels. Thereafter, accelerating declines for gasoline and diesel

consumption more than offset growth in naphtha and jet fuel. Concurrent increases in NGLs and biofuels add further pressure to the demand for refined products, reducing it to 85.9 mb/d by 2030 for a net cumulative gain of just 250 kb/d over the outlook period.

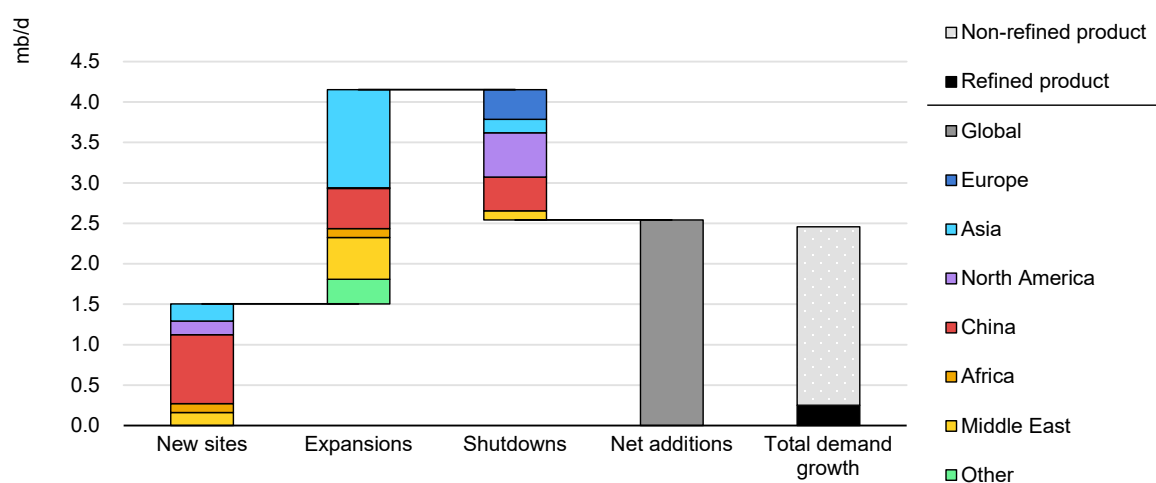
Oil demand and call on refined products (mb/d), 2024-2030

	2024	2025	2026	2027	2028	2029	2030	2024-30 growth
Total liquids demand	103.0	103.8	104.5	105.1	105.4	105.6	105.5	2.5
Biofuels	3.4	3.5	3.7	3.8	3.9	4.0	4.1	0.7
Total Oil demand	99.6	100.3	100.8	101.4	101.5	101.5	101.4	1.8
CTL/GTL*/additives	0.8	0.8	0.8	0.9	0.9	0.9	0.9	0.2
Direct use of crude oil	1.0	1.0	0.9	0.8	0.7	0.6	0.5	-0.5
Total call on oil products	97.9	98.4	99.1	99.7	99.9	100.1	100.0	2.1
Fractionation products**	12.2	12.6	12.9	13.4	13.7	13.9	14.1	1.9
Refined product demand	85.6	85.8	86.2	86.3	86.2	86.1	85.9	0.3
Refinery market share	83.1%	82.7%	82.5%	82.1%	81.8%	81.6%	81.4%	-1.7%

Notes: *CTL = coal-to-liquids and GTL = gas-to-liquids. **Ethane, LPG and pentanes plus, excluding estimated diluent use in North America.

Despite weak demand projections, 4.2 mb/d of new refining capacity is expected by 2030, partly offset by 1.6 mb/d of closures. Net capacity growth, driven by Asia – especially China and India – will exceed demand, outpacing shutdowns in Europe and the United States. To restore balance, utilisation rates must drop, or closures must accelerate beyond historical levels. Both factors will reshape the global refining industry, with reductions dictated by refinery competitiveness. High-cost regions like Europe and the US West Coast are most likely to see further cuts.

Refinery expansion and closures and demand growth, 2024-2030



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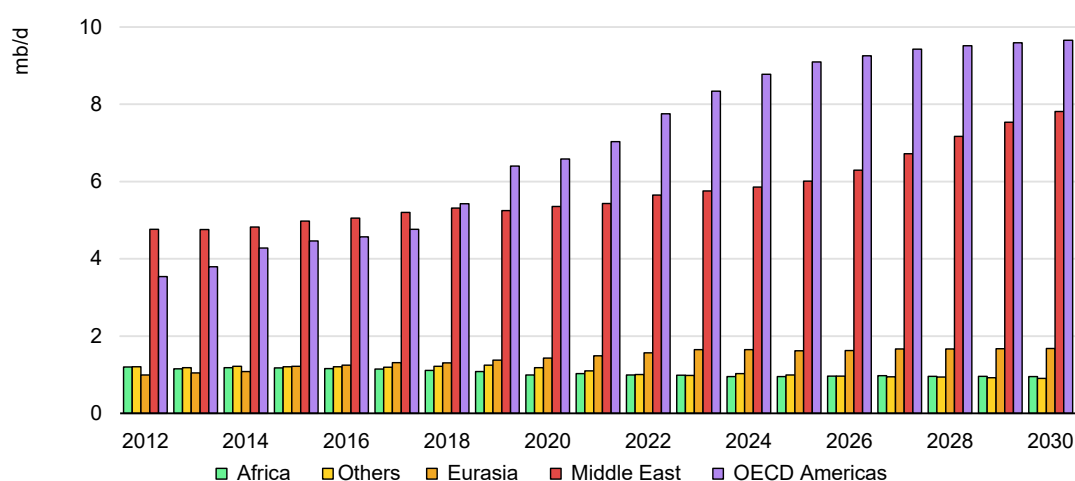
Note: Refined product demand net of CTL/GTL, additives, biofuels, NGLs and direct use of crude.

Shale revolution has upended crude and refining markets

The stellar rise of shale oil and NGLs production from the United States has been the defining characteristic of crude and refining markets since the Global Financial Crisis (GFC) in 2008. The addition of nearly 10 mb/d of light tight oil (LTO) over the last 15 years has redrawn global crude flows and challenged OPEC member states to restrict supplies to support crude pricing. Consequently, shale oil competition has limited the availability of heavy sour Middle Eastern crude grades and thereby compressed upgrading spreads. This shift has provided relatively unsophisticated refineries with a lifeline of cheap, low-sulphur crude that yields a high proportion of light and middle distillates and blend stocks for very low sulphur fuel oil (VLSFO) production. By 2030, US shale oil output growth is forecast to slow to a crawl, with increased Canadian, Brazilian and Guyanese production expected to flip the marginal barrel back closer to the longer-term trend of medium sweet or sour crudes.

However, burgeoning NGLs output, mainly from the United States, has marginalised the role of refineries in meeting petrochemical feedstock demand. Prior to the shale revolution, refinery-supplied naphtha was the dominant source, accounting for 57% of all petrochemical feedstock in 2010. By 2024, this share had slipped to 45%, with ex-refinery naphtha intake largely unchanged in absolute terms over the past 15 years, despite robust growth in petrochemical production. NGLs growth is forecast to remain strong through 2030, adding 1.9 mb/d, split equally between North America and the Middle East.

Regional NGL fractionation supply, 2012-2030



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Note: NGL fractionation volumes net of upstream blending requirements.

On balance, prospects for the refining industry for the 2025-30 period are broadly unchanged from last year's Report. In part, this is due to higher NGLs production

for both the baseline and future growth, even though we have increased our assessment of the share used as upstream diluent. Despite projected demand growth for gasoline having improved, the diesel forecast is now materially lower. With average refinery utilisation rates set to fall in the coming years, the [investment](#) slowdown in new refining capacity seems reasonable.

Precisely which refineries will be most impacted remains unclear but high-cost operations in declining demand centres still appear most at risk. Add to this the shifting mix in consumption, as growing naphtha use contrasts with the contraction in gasoline by 2030, and the burden of adjustment looks likely to fall largely on fluid catalytic cracker (FCC) refineries.

Refining profitability – healthy, but challenges ahead

Refining profitability, though weaker than a year ago, appears to have stabilised at above the mid-cycle conditions of the second half of the last decade. Both structural and cyclical factors contribute to this positive margin environment.

The refining industry is and will remain highly competitive. Refineries benefit from significant economies of scale, but generally have access to standardised technologies and processes. Barriers to entry remain high due to heavy capital investment requirements for new refineries and long project lead times. This is further complicated by stringent planning and environmental regulations that stymie additions in many developed economies and reinforce the dominance of established operators. Competitive advantage can be sustained through the development of proprietary technologies, or licensing them from processing technology suppliers (and potentially competitors) for a fee.

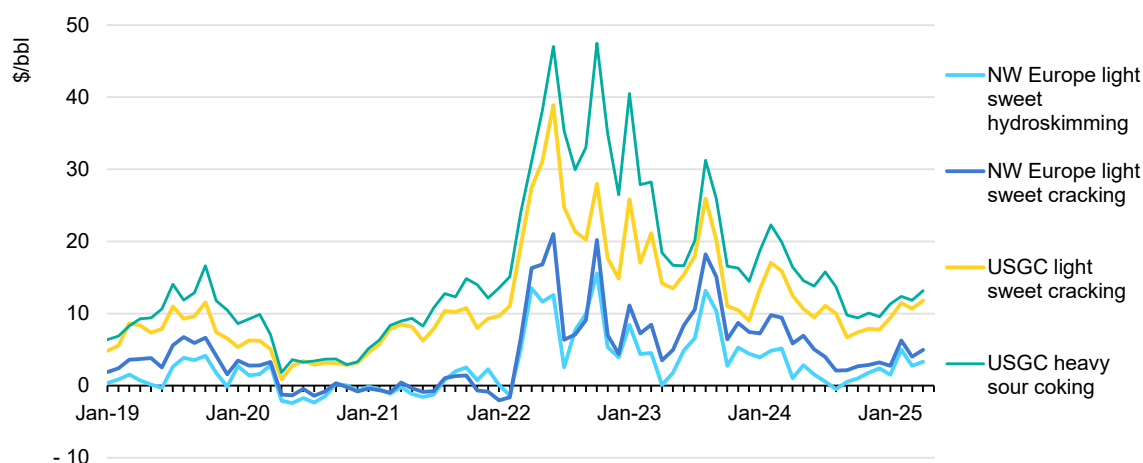
However, refiners often see an incentive to boost runs where additional crude supplies are available. The disparity between the individual incentive to run an additional barrel of discounted crude and the margin implications for the wider market almost always leads to cheap and/or discounted crude supplies undermining industry-wide margins over time.

Historically, more complex refineries have earned higher margins than less sophisticated plants by processing cheaper, heavy/sour crudes or unfinished product feedstocks and converting a greater percentage into premium products such as gasoline, jet fuel and diesel rather than lower value or unfinished grades, including fuel oil. Similarly, refineries in the US Gulf Coast (USGC) or the Middle East with access to cost-advantaged crude, feedstocks or energy have benefited versus higher cost areas such as the North Sea or Asia.

Since the start of the decade, the 22-country OPEC+ group appears to have adopted a production policy aimed at tightening crude markets to support its members' price and revenue aspirations. One tactic to achieve this has been for

OPEC+ members to pivot supplies to Asia, rather than the Atlantic Basin, with Middle East crude exports to the United States falling to a 40-year low in 2024 at just 530 kb/d. OECD Europe crude imports from the Middle East exhibit a similar trend. This has reduced competition between Canadian heavy sour crude grades such as Western Canada Select (WCS) and Middle Eastern grades and tightened heavy crude supply to the USGC.

US and European refinery margins, 2019-2025



Source: IEA analysis based on prices from Argus Media Group, All rights reserved.

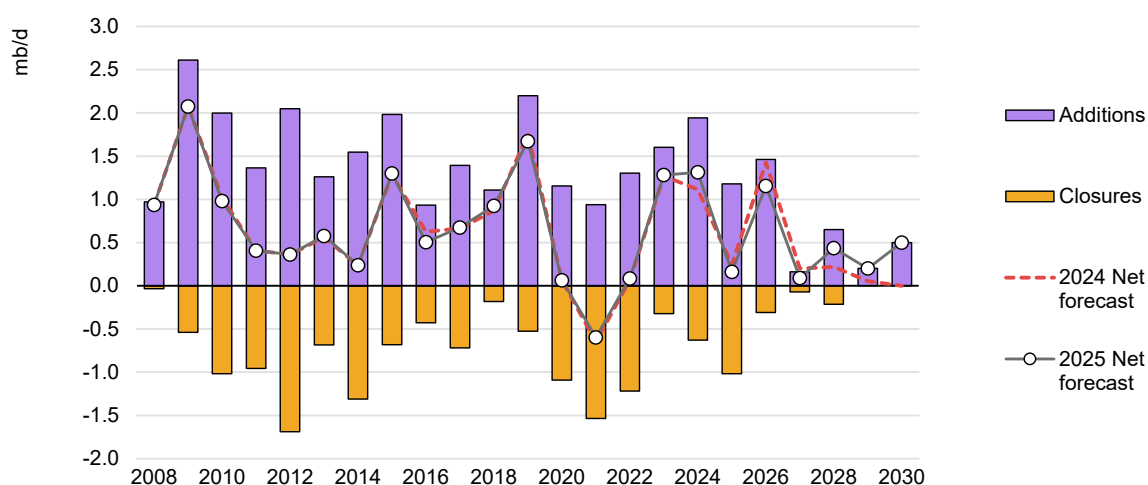
Faced with tight heavy sour crude markets, refineries have had to choose among one of three strategies when optimising processing rates. One option could be to pay up for the spot supplies, thereby sustaining the premium for prompt barrels. Another strategy could be to draw down crude inventories, which from a cash flow perspective might be preferable, but effectively adds to crude market tightness. Lastly, if the margin environment was unfavourable, refineries could cut runs. Viewed through the lens of multi-period game theory, these choices transmit crude market tightness or weakness into the product market.

Hence, crude market backwardation would support product market backwardation where there are limited sources of competing supply. Similarly, widening contango crude market structure would seep into product markets. Consequently, OPEC+ production restraint thus far this decade has been instrumental in supporting the overall level of refinery profits. However, concurrently this restraint has limited the incremental margin available to complex refining versus processing light sweet crude in less sophisticated plants. Moreover, in recent months OPEC+ members have shifted strategy to loosen their production policy and accelerate the reversal of voluntary production cuts. This raises the risk of an oversupplied crude market that could translate into weaker margins if refiners cannot resist the temptation to boost runs with cheaper oil.

Refinery capacity rationalisation needed, but who moves first?

Limited global demand growth for refined fuels and a continued increase in refining capacity worldwide begs the question of where further closures will come from. Closing capacity in recent years, while the industry earned record profits, was hard to justify. However, if margins deteriorate from current levels, industry participants may yet see the incentive to trim capacity. This *Report's* assessments include only announced closures and thus understate the level of rationalisation needed before 2030.

Annual change in global crude distillation capacity, 2008-2030



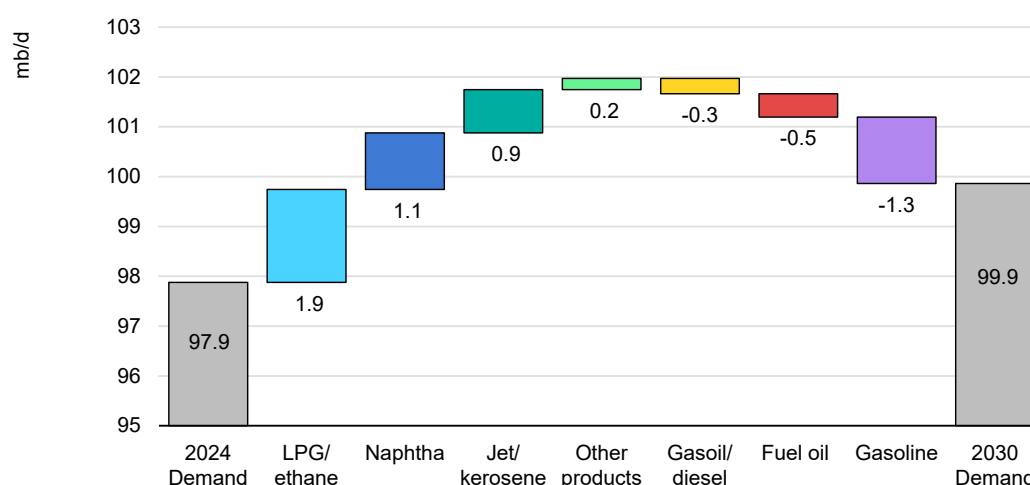
IEA. CC BY 4.0.

Planned maintenance shutdowns are expensive and if the margin environment is unattractive, or declining regional demand boosts export requirements, operators could consider selling or closing an asset. Moreover, if turnarounds entail additional capital expenditures to meet tighter emissions limits, energy efficiency improvements or low carbon fuels processing requirements that individually or collectively generate a negative net present value for such expenditure, then refiners may opt to defer the work, sell the asset or cease operations. While selling refineries that are operating remains a preferred option for many owners, finding prospective buyers is becoming more of a challenge. Excluding the strategic acquisitions of minority stakes by Middle Eastern national oil companies (NOCs) in several Asian world-scale refineries – often tied to long-term crude supply arrangements – foreign NOCs have become increasingly reticent to commit. With the independent refining sector also in an asset-harvesting mode in developed economies, commodities trading houses have become the most obvious candidates. Furthermore, legislation that keeps previous and existing owners liable for future site remediation costs, for example, in California, could also sway the decision.

Adapting to shifting demand trends: The rise of petrochemical value chains

The toughest challenge facing the refining industry is how to adjust to the shifting demand dynamics and continued robust NGLs production growth. Refineries have honed their output yields over time to match regional demand patterns as dictated by price signals. However, the trends detailed in this *Report* will force many refiners to reassess their current configuration and operations. Arguably, refiners will require price signals to provide incentives to adjust.

Oil product demand by major product category, 2024-2030

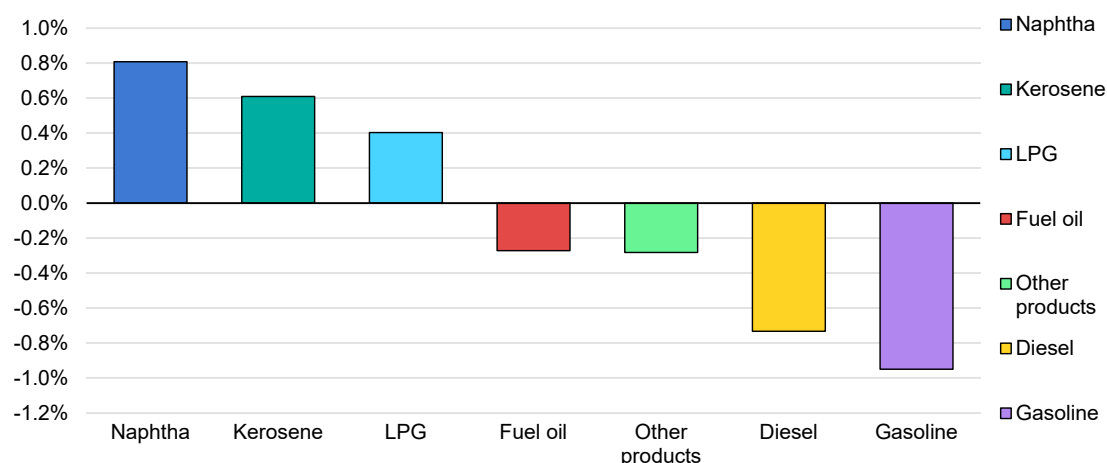


IEA. CC BY 4.0.

Notes: Demand is net of CTL/GTL volumes and biofuels supply. Demand includes volumes met by NGLs supply. Other products demand excludes crude oil used in power generation.

Market pricing incentivises the maximisation of high value, or relatively scarce products such as gasoline or jet fuel, while penalising lower grade products, or those in surplus, including naphtha and fuel oil. These price differentials drive investment in upgrading capacity. Hence, the prospect of a decline in demand for high value products like gasoline and rising requirements for lower value products presents a seemingly intractable problem for the industry. Refiners also face shifting demand trends in the middle distillate market through to 2030, with falling diesel use, accentuated by a 300 kb/d rise in renewable and bio diesel supplies. This is in sharp contrast to continued jet fuel growth. In theory, refineries can adjust crude distillation cut points to shift output between products. However, such changes may only be possible during planned major maintenance shutdowns that occur every four to six years.

Projected global refinery yield percentage point changes, 2024-2030



IEA. CC BY 4.0.

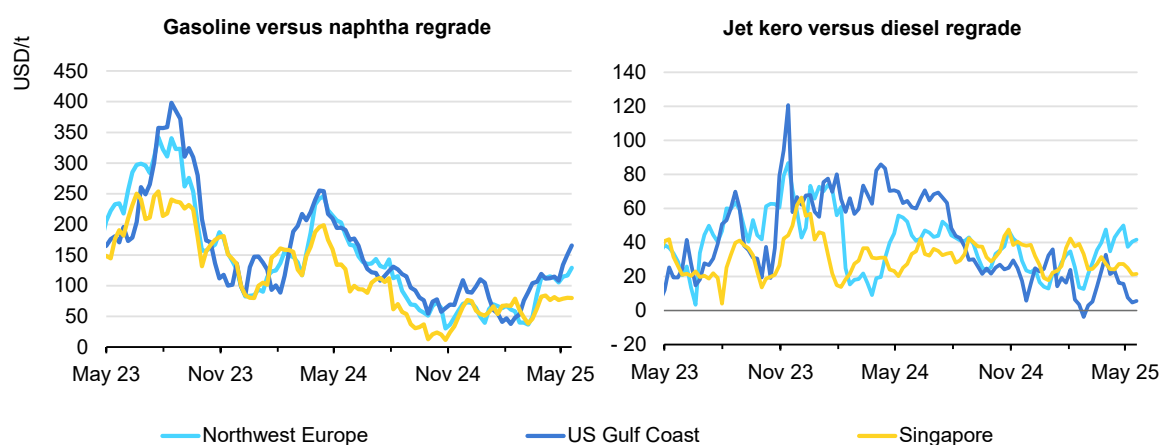
Furthermore, raising jet fuel output may require additional hydrotreating capacity, necessitating further capital expenditure. Similarly, boosting kerosene yields from hydrocracking upgrading units may be possible, subject to similar constraints. Lastly, stringent jet fuel quality specifications may limit the volume available. The price incentive to upgrade diesel to jet fuel varies by region, with European refineries currently seeing the greatest benefit.

The challenge facing refiners in the light distillate market is more complex. The dilemma is how will market participants price the increasing demand for lower-value naphtha and contracting demand for a higher-value gasoline. Moreover, given rising supplies of NGLs-sourced naphtha, refineries do not necessarily set the price of this product. Furthermore, gasoline and naphtha are partially fungible, given that gasoline production typically relies on blending at least six components (including naphtha) to arrive at a finished blend or blend stock. Lastly, not all naphtha is suitable for petrochemical feedstock use. Lighter, paraffinic naphtha is typically used as a feedstock for manufacturing olefins, such as ethylene. However, it can also be processed in isomerisation units to produce a gasoline blending component. Conversely, heavy aromatic naphtha can be reformed to produce high octane gasoline blending component, albeit this contains aromatic chemicals. Incentivising refineries to lower gasoline yields and raise naphtha yields requires a substantial adjustment to current price signals. Effectively, product prices need to dissuade refiners from upgrading low value naphtha into higher value gasoline; something that currently is not happening.

Investment in existing FCC refineries in Europe and the United States to boost output of inherently lower value petrochemical feedstocks appears unlikely. Nor

does the prospect of shuttering FCC or hydrocracking units seem feasible, given the poor profitability of hydroskimming refineries.

Regional product price developments, May 2023-May 2025



Source: IEA analysis based on prices from Argus Media Group, All rights reserved.

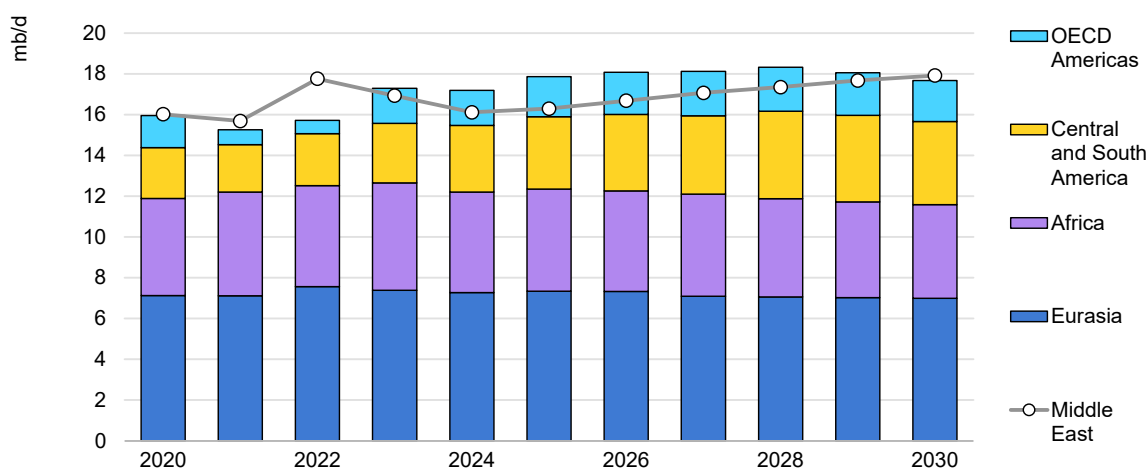
However, new refining projects in China, India and the Middle East are focused on elevating the output of petrochemical feedstocks at the expense of transport fuels. In addition, given that relatively high cost smaller European and US refineries may inevitably be forced to close, an increasing share of crude is expected to be processed in refineries that are specifically designed to maximise petrochemical feedstocks yields, commonly referred to as crude-to-chemicals (CTC) integrated refining and petrochemical hubs. While the profitability of the stand-alone CTC refineries will be below a comparable size fuels refinery, the lower margins achieved on the refining operation will, in theory, be more than compensated for by the integration with massive downstream petrochemical units. Several examples of these refineries are planned or under construction in China and we have adjusted our forecast refinery yields accordingly.

Global crude and refined product trade

Global crude trade will continue to be dominated by the Middle East, with exports from the region increasing from 16.1 mb/d in 2024 to 17.9 mb/d by 2030. This is equivalent to the combined exports from the Americas, Africa and Eurasia. Higher Americas crude production will push additional barrels from the Atlantic Basin into markets East of Suez that are short of crude. Brazilian and Guyanese crude exports will increasingly head east, as 800 kb/d of additional supplies boost Central and South America's crude exports to 4 mb/d by 2030. This follows on from the increase in North American exports to Asia in recent years, first of US LTO and more recently of Canadian heavy crudes following the expansion of the

Trans Mountain Expansion (TMX) pipeline. Atlantic Basin import hubs, e.g. Europe and the US Gulf Coast will compete for crude supplies in the short term, before weaker local demand dampens refinery activity.

Regional crude export flows, 2020-2030



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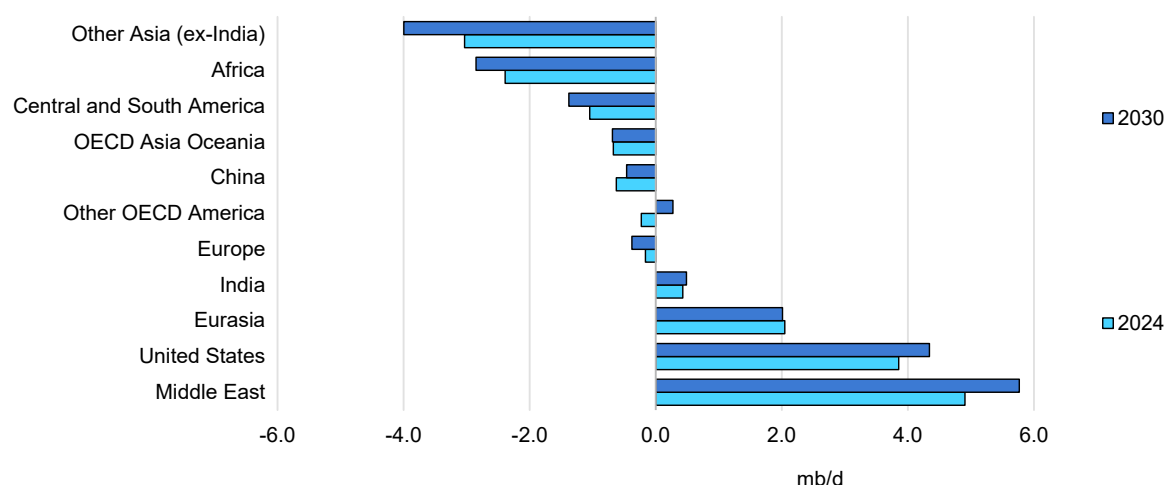
Note: Crude export flows are net of intra-regional flows.

The Middle East is set to lead regional product export gains over the balance of the decade, driven by higher LPG, naphtha and fuel oil volumes. In part, this reflects Qatar's liquified natural gas (LNG) expansions and increased Saudi Arabian non-conventional gas production. Moreover, we expect Middle East refineries to push jet fuel output higher at the expense of diesel, in line with the global trend. Rising fuel oil exports are underpinned by the region's shift towards natural gas in the power sector.

More broadly, several countries that have driven much of the recent rapid expansion in product exports will see only limited progress. In India, rising domestic demand caps export gains, while in the United States falling crude runs reduce surplus product output. Chinese refined product exports are assumed to remain at close to current levels of 40 Mt per annum. However, this could be challenged either by the country's push to cut carbon emissions, or conversely boosted if the government were to yield to pressure from the refining industry to support economic activity.

Jet fuel is set to face the tightest supply conditions over the balance of the decade. Continued jet fuel demand growth will require refiners to boost yields in the face of flat or falling crude runs. Regional export hubs, such as the Middle East, Eurasia and India will see strong demand from regions with continued import needs, such as Europe, Africa and Latin America.

Regional product trade, 2024-2030

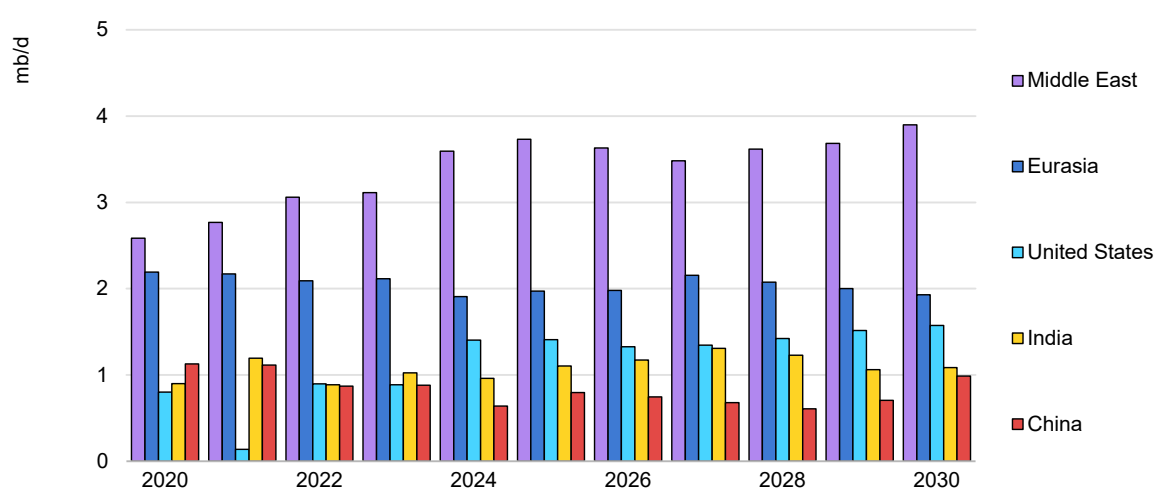


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Note: Regional product trade includes NGLs.

Additional jet fuel supplies will effectively be drawn from the diesel market, which sees falling demand, even though long-term import dependent regions such as Latin America and Africa will continue to need to attract supplies. Conversely, European diesel imports will decline as diesel demand ebbs and despite lower refinery crude processing. Depending on the speed and scale of the adjustment by European refiners, the region's growing jet fuel import requirements could rival or possibly overtake its diesel import needs by the end of the decade.

Major refined product exporters, 2020-2030



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Note: Refined product exports exclude NGLs and biofuels.

Gasoline trade in the Atlantic Basin will see sustained Latin American imports surpassing North American import requirements. European gasoline exports will stabilise in the short term, before pushing higher by the end of the decade as regional demand starts to decline. African gasoline imports should dip as regional crude runs ramp up in the short term, but the continent is expected to see increased product imports from 2026 onwards, as continued strong demand once again outpaces crude processing growth.

How can refiners navigate the challenges ahead?

Global crude distillation unit (CDU) capacity has grown by 13.5 mb/d since 2006, mostly driven by China (+8.4 mb/d), the Middle East (+4.6 mb/d) and India (+2.7 mb/d), while OECD countries decreased by 5.4 mb/d. This regional variation highlights the divide between OECD and non-OECD countries where markets, margins and macroeconomic incentives differ for refiners. Regional disparities are set to continue through 2030 as capacity continues to contract in the OECD (-730 kb/d) but expands by 3.3 mb/d in the rest of the world, led by China (+870 kb/d) and India (+960 kb/d).

The long-term drivers of growth are shifting. Gasoline demand (net of biofuels) is already close to its global pinnacle this year. OECD demand peaked last decade, pulled lower by OECD Asia Oceania and OECD Americas demand contracting since 2016-17, while OECD Europe is forecast to plateau through 2027 and then decline. Similarly Chinese gasoline demand is now falling from its post-Covid highs. Diesel demand in all three OECD regions and China, which accounts for more than 50% of the global total, is also contracting. Conversely, jet fuel demand will continue to grow through the end of the decade. Against this shifting demand outlook, key variables such as energy cost, feedstock accessibility, tighter product specifications, environmental regulations, alternative mobility solutions, and the energy mix in both transport and power generation will define the future of refineries.

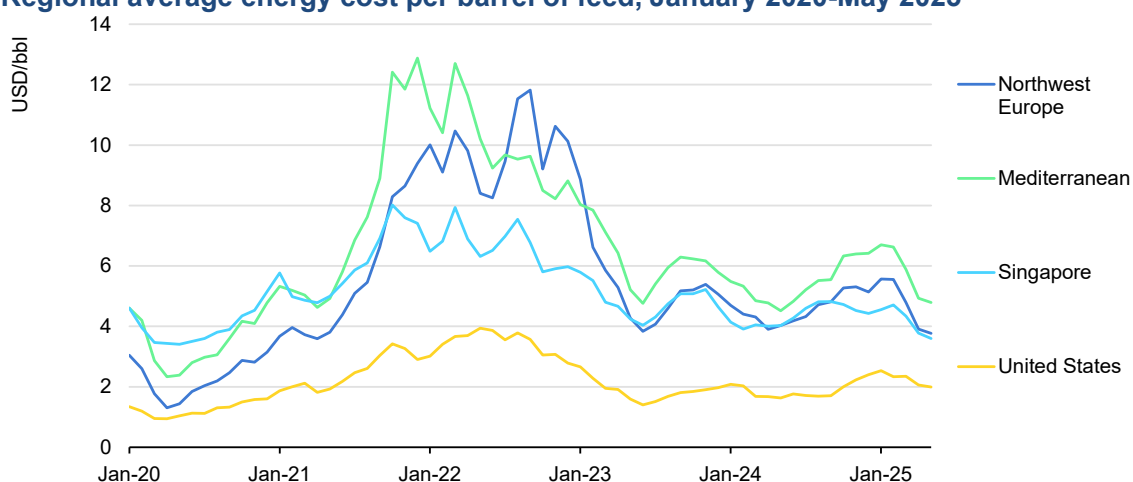
Near term, to survive in a competitive market with surplus capacity, individual refineries and the industry more broadly will have to continuously reassess investment strategies to maintain a healthy margin. A refiner can adapt in the medium term in many ways, but not all are economically viable, depending on the region and market. For example, major maintenance turnarounds (TAR) occur every four to six years and are essential for extending the refinery's operational life. TARs are planned years in advance, and the refinery's management prioritises critical investment areas such as repairs, new product specifications and renewing or upgrading units. To operate, a refiner needs the approval of the government. Consequently, a portion of the investment usually targets improving environmental safety and reducing emissions. In the European Union refiners need to ensure that safety standards meet the Seveso 3 Directive (high risk

industries safety obligations) and environmental legislation. A TAR costs a minimum of USD 50 million and frequently several hundred million dollars. Whether refiners will recoup a return on this investment before the next TAR is a major budgeting challenge.

Focus on efficiency, efficiency and efficiency

Energy efficiency improvements remain a key source of optimisation gains. Energy costs represent about 30-50% of the total operational expenditures. Disparities in energy costs and accessibility impact refinery configurations in each region. The 2022 European natural gas price spike following Russia's invasion of Ukraine boosted the average energy cost to an average of USD 9.90/bbl in Northwest Europe (NWE) as compared to USD 3.50/bbl in the United States. Consequently, EU and OECD Asia refiners have invested in energy efficiency and consolidated their existing assets, switching from fuel oil to natural gas boilers and worked to cut energy intensity at the unit level. For the last two decades, benchmarking has been an essential exercise to improve process efficiency.

Regional average energy cost per barrel of feed, January 2020-May 2025



IEA. CC BY 4.0.

Note: The calculations consider natural gas, LPGs, ethane, fuel oil, electricity, imported heat, petroleum coke consumption and CO₂ credits as specified in the [IEA's refining margins methodology](#).

How European refiners can boost petrochemical yields

Adapting to the shift in global demand trends will require refiners to rapidly change output yields that have been largely enshrined for decades. One strategy adopted by refiners involves modifying the FCC operations and output. The FCC is used to convert heavy distillation cuts – typically vacuum gasoil, but also atmospheric residue – into lighter and higher value gasoline, gasoil, gases and propylene. The unit is key to boosting a refinery's overall margin and is central to the internal flows in the plant. An FCC can be switched from its “conventional” configuration that

maximises gasoline output (or gasoil in the case of European markets) to a high severity set-up that raises production of petrochemical feedstocks. This mode of operation requires increased temperatures and different catalysts to treble the output of propylene. However, this operational switch increases energy costs and generates more coke on the catalyst and therefore more CO₂ emissions.

If Northwest European refiners were facing a situation like China's – namely higher demand for petrochemical feedstocks, lower gasoline consumption and the consideration of carbon credits – would it be more advantageous today to increase propylene production at the expense of gasoline? Over the past five years, the shift to a propylene-max mode of operation would have typically made an average of USD 2/bbl of feedstock but at times during that period – notably post the Russian invasion of Ukraine – it would have cost a refinery as much as USD 12/bbl of feedstock. It is also a question of scale. On average, European refineries have FCC capacity equal to 20% of installed distillation capacity. Consequently, the 10% yield shift in FCC output to propylene would only equate to a 2% change in overall yields.

Northwest Europe high severity FCC margin uplift, January 2020-May 2025



Note: Margin uplift to high severity configuration compared to conventional configuration for NWE FCCs in USD/bbl of feed processed.

Source: IEA analysis based on prices from Argus Media Group, All rights reserved.

Nevertheless, such optimisation may offer short-term relief to refiners if demand for refinery grade propylene is sufficient to absorb additional output. Longer term, as European gasoline demand begins to fall and considering that the current EU propylene market is already long, the future of FCCs in Europe may be in danger, with further closures likely.

Adapting to declining regional demand

Alternatively, the same feedstocks that feed FCCs can be used in a hydrocracking unit (HCK) to produce much higher yields of middle distillates, such as diesel and kerosene. Compared to FCC units, which upgrade through a process of carbon rejection, HCK units hydrogenate and crack feedstock to produce a greater share of, and better quality, middle distillates than FCCs. However, diesel demand in OECD Europe and the United States will decrease by 560 kb/d and 180 kb/d, respectively by 2030, underscoring the need for refiners to adapt their HCK units.

Changing the catalyst and severity of operation and hydrogenation to boost kerosene output remains an option, subject to meeting jet fuel's tight quality specifications. However, even though we anticipate Europe will remain a net importer of diesel, these solutions will fail to match the rapid demand contraction expected by the end of the decade. Consequently, not all European refineries will necessarily be able to operate their hydrocracking units profitably.

Given that a refinery is a heavily interconnected chain of processing units, any change to a unit inevitably affects the overall output. For example, both HCKs and FCCs produce iso-butane and butylene, which serve as feedstocks for the alkylation unit that produces a gasoline blending component (alkylate) that is low aromatic, low volatility and high-octane. This high quality component is crucial to meet current tight quality specifications. Therefore, any reduction in capacity or performance can limit feedstock availability to the alkylation unit, potentially complicating gasoline blending.

Despite the efforts of refiners to remain competitive and viable, pressure is mounting on operations in Europe, OECD Asia Oceania, and more recently the United States. In the 2024-26 period, cumulative refinery closures for these regions total 1.3 mb/d. We only include announced closures in our forecasts, but lower regional processing rates in this *Report* reflect the risk that further closures are likely. The European Fuel Manufacturers Association's (CONCAWE) recent study on "[The potential evolution of Refining and Liquid Fuels production in Europe](#)" suggests a reduction of 2.1 mb/d of CDU capacity in Europe by 2030 is possible.

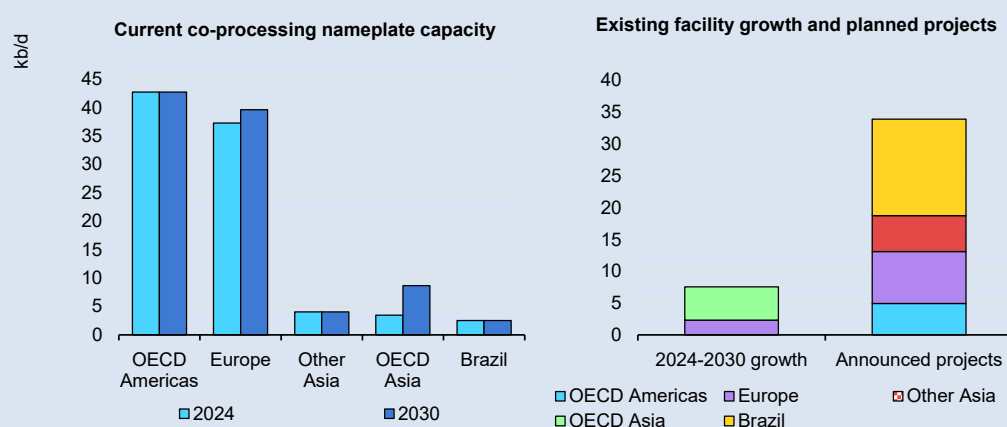
Biofuels co-processing expanding

During the last decade, stronger legislation on biofuel blending limits and better economic incentives supported by higher subsidies and prices have pushed refiners to process vegetable oils, waste oils and fats in hydrotreating units (HDT) to produce hydrotreated vegetable oils (HVO) and, more recently, sustainable aviation fuel (SAF).

Processing biogenic feedstocks to make HVO has the advantage of producing a fuel with similar properties to diesel. HVO, which can be blended up to 100%, is a good alternative to biodiesel (fatty acid methyl ester or FAME) that has lower blending limits. A recent trend in the refining landscape is the co-processing of renewable feedstocks with conventional crude and feedstocks. Incorporating a 10-20% vegetable oil feedstock to existing hydrotreating units (HDTs), and to a lesser extent to hydrocrackers (HCKs), can scale-up HVO/SAF production while saving capex compared to building new stand-alone plants. In the case of a unit at 100% utilisation rate, the refiner must reduce the intake of feedstocks derived from crude oil to allow for the addition of vegetable oil to the HDT or HCK. One of the main constraints to produce HVO/SAF is the higher hydrogen consumption from renewable feedstock, due to its oxygen content that is removed via hydrogenation. Other co-processing pathways are under development, for example the use of FCCs that do not consume hydrogen, to treat biomass-derived feedstocks such as pyrolysis oil or Fischer-Tropsch waxes

By 2030, expansions at existing sites could increase global co-processing capacity by 7 kb/d from the 2024 baseline of 90 kb/d. A further 34 kb/d of proposed projects – if realised – could lift capacity to 130 kb/d. Currently SAF and HVO remain the most economically interesting biofuels to produce. In 2024, co-processing 10% of used cooking oil (UCO) with 90% vacuum gasoil (VGO) was profitable for NWE hydrocrackers. This presents a good opportunity for European refiners, even as diesel demand is declining, with hydrocrackers having the capability to maximise kerosene production over diesel.

Refinery co-processing capacity and additions, 2024-2030



Source: IEA analysis based on data from *Argus Media Group*, All rights reserved.

However, in both cases vegetable oils need an extra pre-processing treatment to remove impurities that can affect the reliability of the upgrading unit, and to prevent catalyst deactivation and pipe corrosion. A hydrogen-long refinery will prefer hydrotreating units to maximise HVOs or SAF production over FCC and HCK

co-processing where biogenic feedstocks are limited by availability or technological bottlenecks.

There is a clear trend to maximise the use of residue and waste oils because they have low emission intensity and meet feedstock requirements in the European Union. The high demand for these feedstocks resulted in a surge of UCO costs in 2025, pushing processing margins negative in the United States in April 2025. Further legislation to increase the ratio of biofuel products to finished products is planned in the coming years. For example, the European Union has set a mandate for 6% SAF blending by 2030.

Refining capacity

Global refining distillation capacity (including both crude distillation units and condensate splitters) is on track for a net expansion of 2.5 mb/d between 2024 and 2030. Approximately 4.2 mb/d of new and expanded capacity is scheduled to come online, offset by around 1.6 mb/d of announced closures. Consequently, refining capacity is forecast to reach 108.3 mb/d by 2030. However, the pace of growth is significantly slower than the historical trend, with the five-year average annual net additions at just over 400 kb/d. This represents half of the average for the post-GFC decade of nearly 800 kb/d. The striking slowdown in demand growth for refined products weighs heavily on investments in refining capacity.

Regional refinery capacity and utilisation, 2024-2030

	2024	2030	Change	2024	2030	Change	2024	2030
	Total capacity (mb/d)			Refinery throughput (mb/d)			Utilisation rates	
OECD Americas	22.2	21.8	-0.4	19.1	18.6	-0.5	86%	85%
United States	18.3	17.7	-0.5	16.1	15.5	-0.6	88%	87%
C and S America	5.7	5.9	0.1	3.7	3.8	0.0	64%	64%
OECD Europe	13.7	13.3	-0.4	11.2	10.3	-0.8	82%	78%
Other Europe	0.8	0.8	0.0	0.5	0.4	0.0	57%	53%
Africa	3.7	4.0	0.2	1.8	2.2	0.4	47%	55%
Eurasia	9.2	9.3	0.2	5.7	6.0	0.3	62%	64%
Middle East	11.6	12.2	0.6	9.8	10.0	0.3	84%	82%
OECD Asia	7.2	7.2	0.0	5.6	5.3	-0.3	78%	74%
China	18.5	19.5	0.9	14.6	14.9	0.2	79%	76%
Other Asia	13.1	14.3	1.3	10.8	11.8	1.0	82%	82%
India	5.8	6.8	1.0	5.4	6.3	0.9	92%	92%
World	105.7	108.3	2.5	82.6	83.3	0.6	78%	77%
Atlantic Basin	53.8	53.6	-0.2	41.1	40.5	-0.7	76%	76%
East of Suez	51.9	54.7	2.7	41.5	42.8	1.3	80%	78%

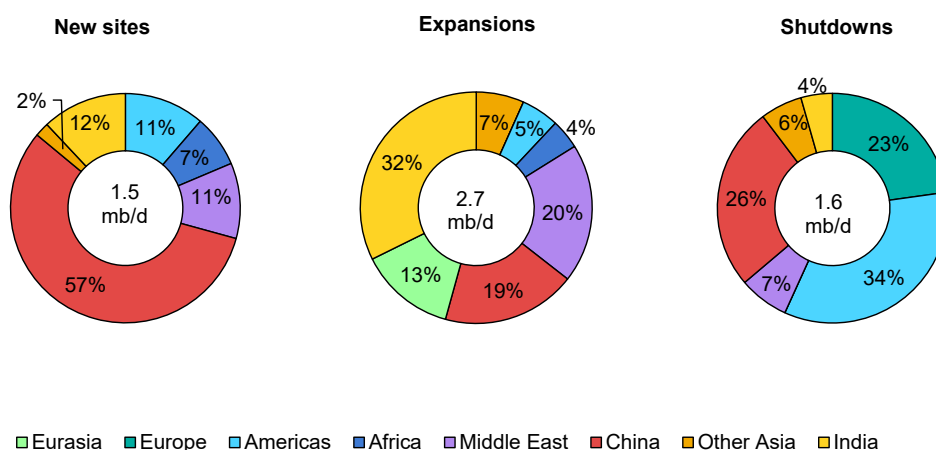
Investments continue to diverge regionally through the end of the decade. Atlantic Basin refining capacity will remain in structural decline, driven by transportation decarbonisation policies and mounting cost pressures. Conversely, countries East of Suez continue to invest in both world-scale petrochemical-oriented refineries

and smaller domestic-focused plants. Consequently, the refining system is entering a new phase – defined not only by expansion, but by adaptation. By 2026 East of Suez refining capacity will exceed the Atlantic Basin for the first time.

Announced shutdowns total 1.6 mb/d for 2025-30, equivalent to an annual decline of around 250 kb/d. Although slower than the average of 800 kb/d over the past decade, we only include announced shutdowns in the forecast, as refiners typically confirm closure plans 12 to 18 months in advance. As such, there is potential for substantial additional closures in Europe, North America and China and we have adjusted utilisation rates lower for heavily exposed regions in anticipation of further capacity rationalisation. In 2025, over 1 mb/d of capacity will be closed – making it the largest single year of rationalisation since 2022. Most of this is concentrated in OECD markets. The United States accounts for over 400 kb/d of closures, followed by Europe with 370 kb/d.

Conversely, 2026 will see nearly 1.5 mb/d of gross capacity additions, driven overwhelmingly by India, China and the Middle East, with announced closures totalling only 300 kb/d. Greenfield investments have all but stalled, with expansions, upgrades and petrochemical integration dominating planned projects – reflecting a shift towards strategies that enhance margins without adding substantial new crude intake. This signals a structural transition in refining – from volumetric growth to value, improved flexibility, emissions mitigation and optimising product yield.

Capacity additions and closures breakdown by region, 2024-2030



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Regions East of Suez remain the anchor of future growth. By 2030, China will retain its status as the world's largest refiner, with capacity rising to 19.5 mb/d,

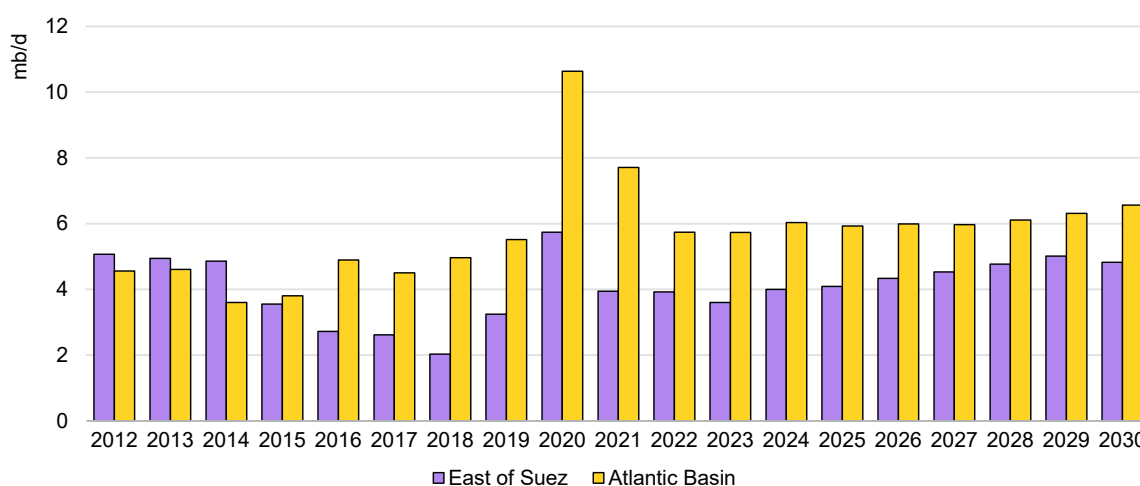
even though excess capacity will remain at nearly 2.2 mb/d. The United States follows at 17.7 mb/d, with India at 6.8 mb/d, just behind Russia with 7.1 mb/d.

Excess refinery capacity – a catalyst for change

Excess capacity has been a long-term feature of the refining industry and looks set to increase by 1.8 mb/d through the forecast period, to 11.4 mb/d by 2030. It is determined by the surplus of installed CDU and condensate capacity, adjusted for average maintenance (which we assume is 14%) above projected refinery runs. Unlike the upstream component of oil markets, neither countries nor companies hold a strategic buffer of refining capacity to stabilise product cracks, preferring to allow international trade to meet local product imbalances.

Conversely, cyclical over-investment – initially in OECD countries and, more recently, in markets such as China – have resulted in a structural surplus. Shifting trends in product demand have exacerbated this build-up of excess capacity, with the highly competitive product markets exposing capacity that is poorly configured to product market needs. Arguably, operational reliability issues, particularly when regional refining systems are running at high levels of utilisation, make the case for an additional 6-8% of marginal capacity that can be brought into service. Nevertheless, growth through 2030 in Asia, the Middle East, and Africa – serving both domestic demand and export markets – will continue to reshape the global downstream map.

Refining industry excess capacity, 2012-2030



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Note: Assumes 14% downtime for maintenance is required.

East of Suez, excess capacity reaches nearly 4.9 mb/d by 2030, 820 kb/d higher than in 2024, with China and Other Asia contributing to most of the rise. In OECD Asia Oceania, excess capacity in both Japan and Korea will increase by 100 kb/d

and 180 kb/d respectively to 500 kb/d due to declining demand. Singapore's excess capacity will reach 220 kb/d, with its long-term viability continuing to depend on export margins.

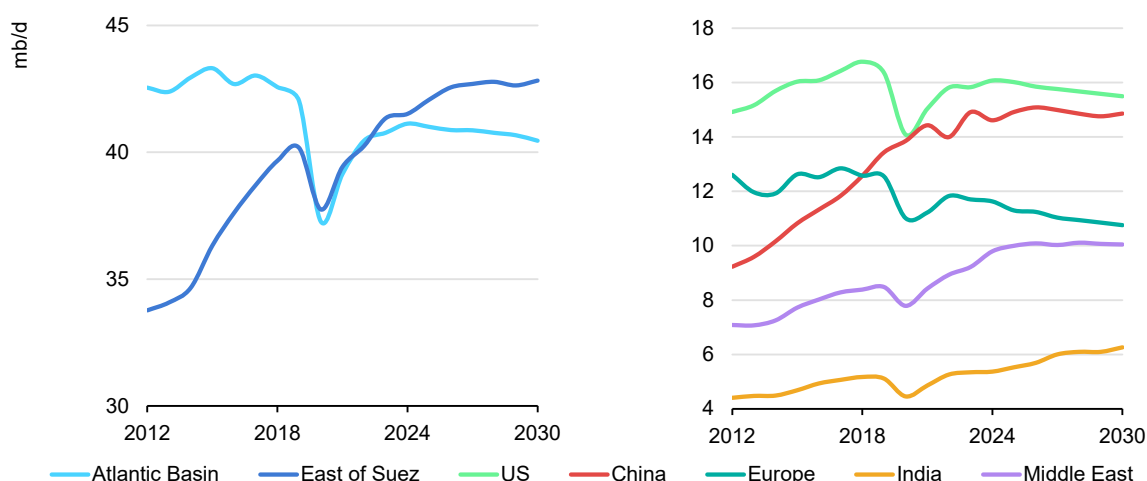
In the Atlantic Basin, excess capacity will grow marginally to 6.5 mb/d by 2030. The Americas will hold steady at 1.7 mb/d while Africa's excess capacity will total 1.4 mb/d, with much of it concentrated in underinvested plants with low-utilisation rates. In OECD Europe, excess capacity will swell by 650 kb/d to 1.6 mb/d, making it one of the most vulnerable regions for capacity rationalisation. In Eurasia, modernisation and a gradual recovery from conflict-related damage keeps excess capacity steady at 400 kb/d, but the region currently still holds a surplus of 2.4 mb/d, with utilisation expected to improve only modestly.

Regional developments

Atlantic Basin

Crude throughput in the Atlantic Basin is expected to fall by 670 kb/d to 40.5 mb/d by 2030, largely on weaker European and US runs. This compares with a modest 210 kb/d decline in capacity, as growth in Africa and Eurasia partially offsets closures in Europe and the United States. Excess capacity remains high, at around 6.5 mb/d, with average utilisation dipping slightly to 76%. Contracting demand in OECD markets like North America and Europe puts older, less efficient assets at risk.

Atlantic Basin versus East of Suez refinery throughputs, 2012-2030



IEA. CC BY 4.0.

Product exports rise by 320 kb/d to 3.5 mb/d, as regional demand for refined products contracts. Import requirements for transport fuels, including gasoline and

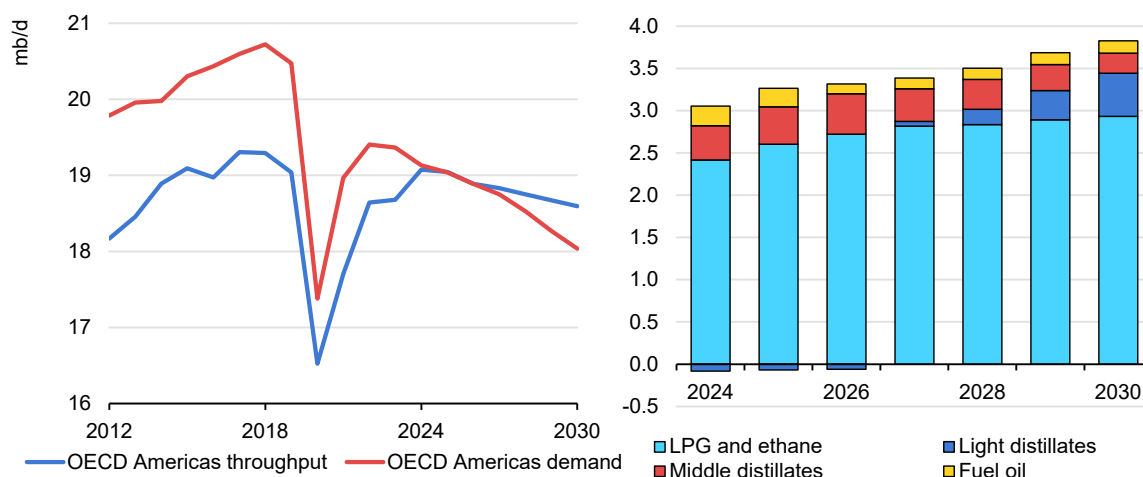
diesel, will decrease by 390 kb/d, as demand drops by 1.5 mb/d due to the expanding EV fleet and increasing vehicle efficiencies. Conversely, ethane and LPG demand grows by 700 kb/d to 8.1 mb/d, however NGLs supply increases faster, resulting in a net 110 kb/d increase in exports to 1.7 mb/d. These volumes will increasingly seek outlets in emerging market petrochemical hubs in Asia and growing LPG demand centres – particularly in Africa – where clean cooking initiatives are accelerating.

Our upstream assessment for the Atlantic Basin sees crude production increasing by 1.1 mb/d to 48.6 mb/d in 2028, before slipping back to 47.7 mb/d by 2030. Consequently, we expect the Atlantic Basin's crude surplus to widen by approximately 870 kb/d to 7.1 mb/d by 2030. Similarly, NGLs supply increases by 860 kb/d to 9.2 mb/d by 2030.

OECD Americas

The OECD Americas faces the challenge of declining regional refined products demand over the outlook period. Consequently, refining throughputs, capacity and utilisation are all forecast to contract by 2030. Average regional capacity utilisation, currently at 86%, dips to 85.2% by 2030, with slightly lower US rates offset by improved Mexican utilisation. As oil consumption growth falters, demand for refined product falls by close to 1.1 mb/d by 2030, significantly outpacing the region's 370 kb/d decline in capacity.

OECD Americas refinery runs, demand and net product balances, 2012-2030



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The **United States** continues to dominate regional crude processing capacity, but the sector has entered a phase of structural contraction. Despite recent expansions (most notably ExxonMobil's 250 kb/d CDU at Beaumont, Texas,

commissioned in early 2023), gains are being overshadowed by a string of closures. In 2025 alone, the United States is set to close over 400 kb/d of capacity. On the Gulf Coast, LyondellBasell's 265 kb/d Houston refinery permanently shut in Q1 2025 after efforts to find a buyer proved unsuccessful.

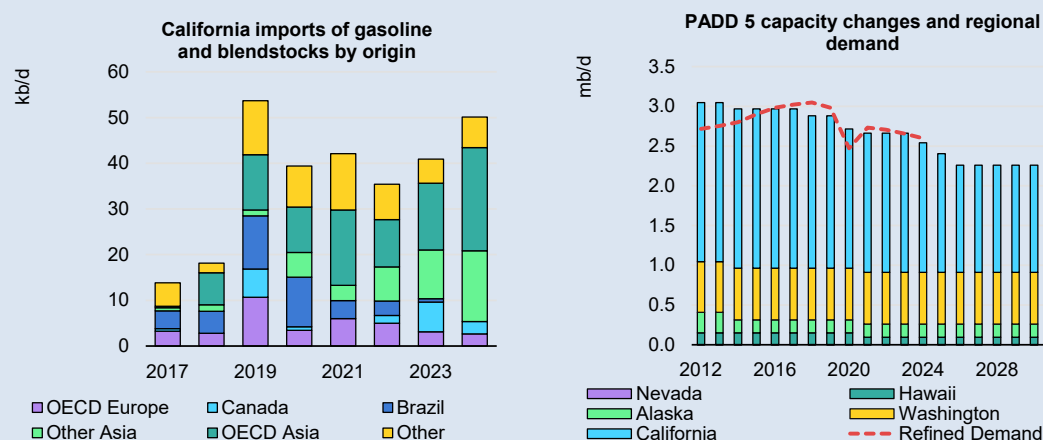
California is undergoing a particularly sharp rationalisation (see *All eyes on West Coast refineries*). Phillips 66 shuttered its 120 kb/d Rodeo refinery in early 2024 and will close its 140 kb/d Wilmington facility in Q4 2025 as part of a broader portfolio optimisation strategy. Further losses are expected next year, with Valero's 145 kb/d Benicia refinery slated for shutdown and repurposing in H1 2026 amid intensifying environmental compliance costs and regulatory pressure. These developments exemplify a broader national trend of refinery attrition that has closed 2.6 mb/d of capacity since 2010. Factors contributing to the decline include ageing infrastructure, higher costs from environmental regulations in many states and rising investment requirements to maintain competitiveness and meet nationally mandated emissions reduction targets.

All eyes on West Coast refineries

Refining capacity on the US West Coast (PADD 5) is undergoing a sustained and significant contraction. Between 2018 and 2026, over 700 kb/d of crude distillation capacity has or will be closed. This represents nearly one-quarter of the region's operational baseline. The wave of PADD 5 closures began with Paramount Petroleum's 85 kb/d refinery and Island Energy's 55 kb/d Honolulu plant in late 2018, and Marathon's 170 kb/d Martinez refinery in 2020. In 2024 Phillips 66's repurposed 120 kb/d Rodeo facility for renewable fuel production. By the end of 2024, cumulative crude processing capacity losses had reached 420 kb/d – more than triple the 130 kb/d decline in regional transport fuel demand over the same period. The 2025 slated closure of Phillips 66's 140 kb/d Wilmington refinery will lift total shutdowns to 560 kb/d. The latest announcement for the 2026 closure of Valero's 145 kb/d Benicia refinery will leave only 12 crude oil refineries in California, with a total capacity of 1.3 mb/d.

These shutdowns are not being driven by poor refining margins – which remain relatively healthy – but by financial and policy challenges, including a cleaner energy landscape that increasingly favours renewable fuels and vehicle electrification. These include stringent environmental compliance costs, extensive permitting hurdles, and unique fuel specifications such as the California Reformulated Gasoline Blendstock for Oxygenate Blending (CARBOB) for gasoline, alongside a progressively restrictive policy framework shaped by the Cap-and-Trade system that limits GHG emissions and the Low Carbon Fuel Standard (LCFS).

US West Coast refined product trade and capacity



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Sources: IEA analysis based on trade data from *Kpler*. PADD 5 West Coast demand data is from the US EIA.

While the long-term shift toward EVs and more efficient internal combustion engine (ICE) powertrains is expected to gradually lower gasoline demand, the current rate of refinery closures is outpacing this transition. The pace of capacity rationalisation keeps the region net short products and requires continued imports, particularly for gasoline. In 2024, California imported nearly 50 kb/d of gasoline and blending components, largely from Asia. However, few refineries globally are configured to easily meet California's strict fuel standards, limiting alternative supply sources. The US Gulf Coast serves as a source of replacement barrels, though high shipping costs associated with the compliance of the Jones Act – which mandates cargo transported by water between US ports must be carried on US flagged, built and owned vessels – may limit arbitrage opportunities unless price differentials widen.

The tightening product balance in California is also expected to ripple into adjacent markets – including Arizona, Nevada and the Pacific Northwest – which rely on inter-regional marine and pipeline flows from the state. This structural rationalisation is likely to increase price volatility and import dependency over the medium term.

Mexico's state-owned company Pemex continues to face difficulties boosting downstream output as it seeks to substitute imports with domestic production of products. The much-delayed 340 kb/d Olmeca refinery in Dos Bocas on the Gulf of Mexico is currently ramping up, and expected to reach full operations in 2026. The first 170 kb/d phase started in late 2024, with the second tranche expected to be running later this year. This will be a major milestone in addition to Pemex's overhaul programme for six refineries, which includes upgrading units at Tula,

Salina Cruz, and Salamanca. Supported by close to USD 8 billion in planned investments, Mexico aims to secure self-sufficiency and retail price stability via increased production of gasoline, diesel and jet fuel. However, if successful by the end of the decade, the expected decline in Mexican crude production will push the country closer to becoming a net crude importer and tighten supplies of heavy sour crude for US Gulf Coast refineries.

Crude and product balances

The net crude export position of OECD Americas is expected to gain 290 kb/d through 2030, to around 2 mb/d. This stability masks underlying structural shifts within the region's supply profile. Canadian heavy crude and diluted bitumen (dilbit) production is set to rise significantly (+470 kb/d), boosted by upstream expansions and improved takeaway capacity, which directs as much as 600 kb/d into Asian markets. However, this gain is more than offset by a 660 kb/d decline in conventional crude and LTO production, driven primarily by declines in Mexico and slowing output growth in the United States.

Net product exports will rise by a significant 990 kb/d to 4.6 mb/d by 2030. More than half of the increase is in LPG and ethane, which combined will reach 2.9 mb/d by the end of the decade, reinforcing the region's role as a key supplier of petrochemical feedstocks. The gasoline/naphtha balance shifts towards being net long by 2030, driven by rising pentane plus/naphtha supply from NGLs. Gasoline net imports shrink by 550 kb/d to just 160 kb/d as domestic demand shifts into structural decline and regional supplies rise. Notably, the launch of Mexico's Dos Bocas refinery will play a critical role in enabling the region to shift to a net exporter position in light ends by 2030. By contrast, middle distillate balances tighten modestly, as available export volumes decline amid refinery closures and continued jet fuel demand growth.

Challenges

The OECD Americas refining sector is navigating a complex and often contradictory landscape. Despite healthy post-pandemic refining margins (averaging 20% above 10-year norms), closures continue as costs of regulation and modernisation outweigh future cashflows. This paradox reflects a shift from current cash-flow-based measures of viability to include future capital investment requirements. Hence, refineries are shutting not on the basis of current profits, but due to the prohibitive cost of modernisation. With US demand for refined gasoline having peaked late last decade, many operators are foregoing major upgrades. This trend favours larger, integrated plants capable of processing a wider crude slate and accessing export markets. By contrast, older, smaller sites – particularly those isolated from logistic hubs or with limited product flexibility – are increasingly vulnerable.

Central and South America

Refining capacity in Central and South America sees minimal change through 2030, rising by just 130 kb/d to 5.9 mb/d. The increase is driven solely by Brazil's phased expansion of its RNEST refinery in Abreu e Lima. A 15 kb/d expansion was completed in Q1 2025, completing the Train 1 modernisation and emissions retrofitting. A more substantial addition of 115 kb/d from Train 2 is scheduled for Q2 2029, which will bring RNEST's total capacity to 245 kb/d. This project, stalled since 2015, has been revived under Petrobras' new downstream strategy that will see domestic crude runs effectively meet product demand.

Refinery throughput in Central and South America increases by 50 kb/d over the forecast, yet at 3.8 mb/d remains well below nameplate capacity, reflecting structural underutilisation. Venezuela accounts for the majority of this surplus, with 1.3 mb/d of installed capacity continuing to operate at less than 25% utilisation due to years of underinvestment, poor maintenance and ongoing sanctions. Similarly, PDVSA's 320 kb/d Isla refinery in Curaçao continues to sit idle. Once considered a key regional supplier, the facility has yet to secure a viable operator since PDVSA ceased operations in 2019. With refined product demand rising by 350 kb/d to 5.2 mb/d by 2030, the lack of regional refining activity reinforces import dependence for product supply.

Crude and product balances

Central and South America's net crude exports growth of nearly 800 kb/d over the forecast period, to 4 mb/d by 2030, is particularly striking. This is the second-largest regional increase globally, after the Middle East. The surge reflects a widening disconnect between upstream supply and refining capacity, with limited investment in domestic refining infrastructure leaving much of the region's output – particularly medium and heavy sweet crudes – destined for export. While the United States continues to absorb a portion of these volumes, the distillate-rich crudes are of particular interest to refineries in China, India and Europe.

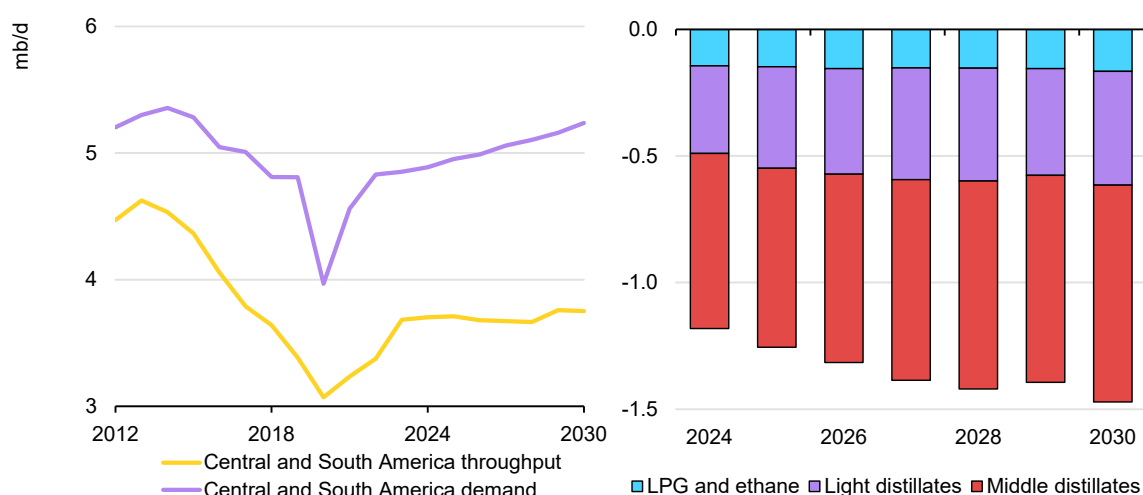
Conversely, product import reliance in Central and South America is set to widen by nearly 330 kb/d to 1.4 mb/d – broadly in line with a 350 kb/d increase in refined product demand. Diesel leads the growth, adding 140 kb/d to import requirements, while gasoline and naphtha together contribute a further 100 kb/d. However, when Mexico's Dos Bocas refinery ramps up, this will boost North American gasoline and diesel exports, freeing up volumes, particularly from the US Gulf Coast to meet the growing Central and South American shortfall.

Challenges

Central and South America faces persistent product supply deficits due to a lack of refining capacity and despite abundant upstream potential. The region is set to

deliver strong production growth in countries like Brazil, Guyana and Suriname. However, political instability and underinvestment continue to weigh heavily on downstream infrastructure. Refinery throughput remains stagnant, with many facilities ageing or idle. Regional demand for refined products is growing, but with limited new investment in domestic capacity, import reliance is rising steadily. Without substantial investment or improved operational performance, the refining sector will remain a bottleneck in an otherwise resource-rich region.

Central and South America refinery runs, demand and net product balances, 2012-2030



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Africa

Africa's refining landscape is undergoing a structural transformation, with major capacity expansions underway and several small to medium-sized projects advancing across key markets. The commissioning of the independent 650 kb/d Dangote refinery in Nigeria marks a turning point for West Africa's product balances. After years of delays, the continent's largest refinery started processing crude in H1 2024. Crude imports averaged around 400 kb/d by Q1 2025, including cargoes from the United States, Brazil and Angola. While operational issues, problems securing domestic crude allocations and delays in the integration of secondary units persist, throughput is expected to stabilise at above 70% of capacity by late 2025. Once fully online, the facility will significantly reduce Nigeria's fuel import needs and reconfigure regional and Atlantic Basin product flows. Overall, Africa's refining capacity is projected to increase by 220 kb/d between 2024 and 2030.

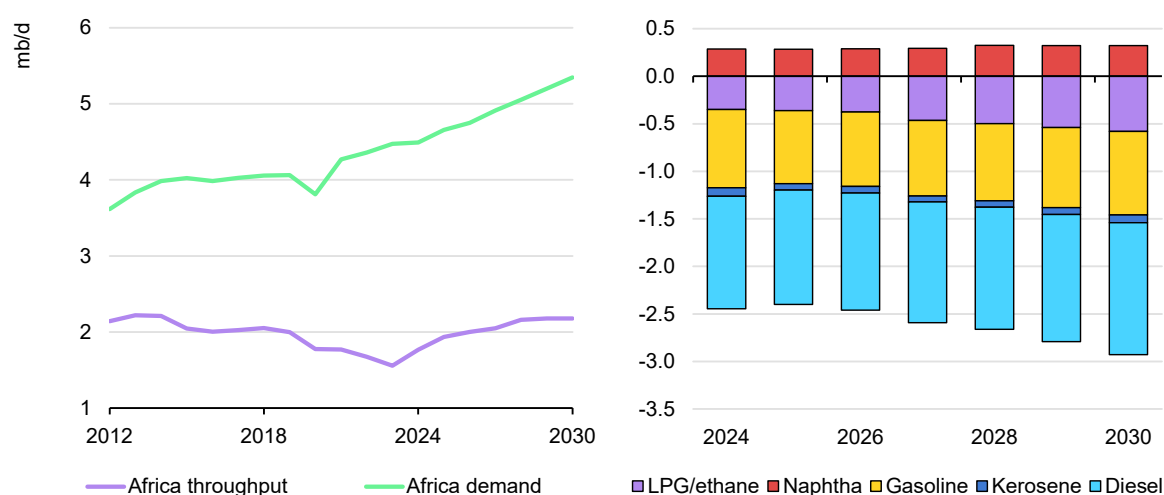
Alongside this flagship project, **Nigeria** is working to revive its state-owned refineries, with limited success thus far. The 210 kb/d Port Harcourt complex saw its older 60 kb/d crude unit restart in December 2024 after more than a decade of

inactivity. However, processing proved short-lived and the refinery is not currently operating. Another 150 kb/d CDU at the same site is under rehabilitation and could potentially come online during 2026. Similarly, the 125 kb/d Warri refinery briefly resumed partial operations in late 2024, but is now back under maintenance pending a potential restart next year. Lastly, the 110 kb/d Kaduna refinery is still slated for rehabilitation, but unlikely to re-enter commercial operations before early 2027 and probably much later. Together, if successful, these efforts could lift Nigeria's total operational capacity towards 1 mb/d by the end of the decade. Elsewhere in West Africa, we have included small domestically-focused projects in Ghana and in Congo, which are progressing after numerous delays.

Algeria's refining capacity will also expand with the construction of a new 110 kb/d refinery at Hassi Messaoud. This much delayed project is now estimated to start-up in 2028. Algeria's older facilities at Skikda, Algiers and Oran were previously modernised to improve product quality.

In East Africa, **Uganda** has progressed plans for a 60 kb/d greenfield refinery to process Lake Albert crude. In April 2025, a lead investor was secured and key agreements on government participation and feedstock supply are being finalised, with commissioning targeted for around 2027. However, in the absence of more concrete construction plans, we have excluded this project from the forecast.

African refinery runs, demand and net product balances, 2012-2030



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Crude and product balances

Africa's net crude exports are set to contract by 350 kb/d to 4.6 mb/d by 2030. This shift is driven primarily by the ramp-up of regional refining capacity – most notably the 650 kb/d Dangote refinery in Nigeria, along with smaller projects in Algeria, Ghana and Congo. These latter facilities are designed to process locally

produced crude, thereby reducing the region's exportable surplus. Thus far, higher African crude runs have been met by increased imports, as the Dangote refinery has focused on acquiring and processing the most cost effective grades and not simply acquire relatively expensive domestic crude cargoes.

Despite ambitious plans to expand domestic refining capacity, regional demand growth will outpace product supply. Strong demand growth of 860 kb/d, of which 630 kb/d is for refined fuels and 230 kb/d is LPG, will lift the continent's product imports requirement significantly. The region's refined product deficit is projected to widen by 230 kb/d to 2.3 mb/d by 2030. Gasoline and diesel will remain the products with the largest shortfalls, with diesel alone accounting for 1.4 mb/d. Africa's 320 kb/d surplus of naphtha is insufficient to offset the broader product import requirements. Furthermore, the ramp-up of multiple clean cooking initiatives have led to a large demand increase for LPG, with deficits set to nearly double over the forecast period to 570 kb/d.

Challenges

Africa is losing ground in its fight to move towards greater refined products self-sufficiency. Despite refining sector expansions, with several large projects underpinned by national energy security strategies, demand growth will outpace domestic refined product supply. Furthermore, implementation risks remain high, and even if the net forecast additions – alongside the ramp-up of large already-commissioned projects – are delivered, Africa will still only be a marginal contributor to global refining growth in the coming decade. Without substantial acceleration in refinery projects or improved utilisation rates, Africa will continue to rely heavily on refined product imports to meet rising demand, especially for transport fuels.

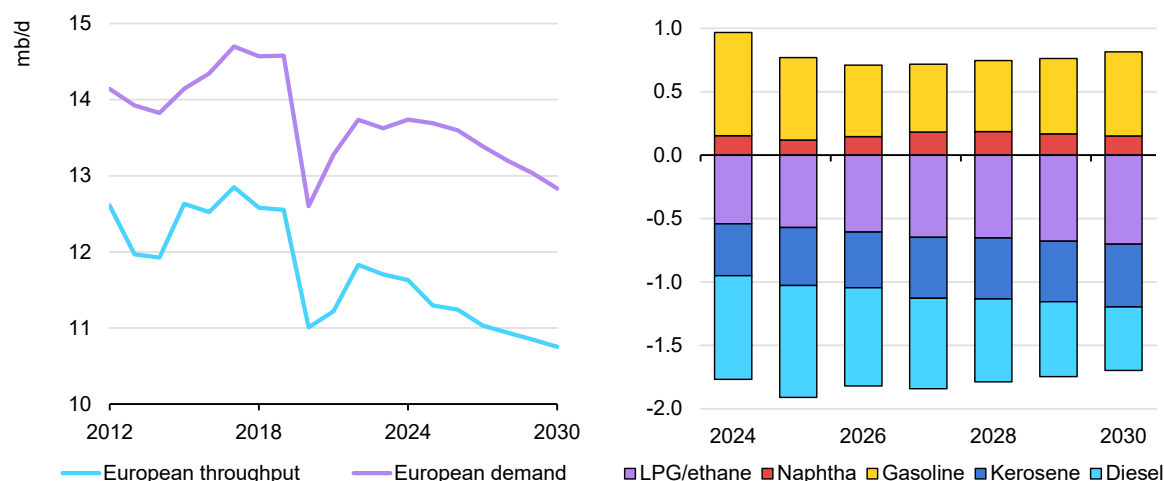
Europe

Total product demand in Europe, (including OECD and non-OECD Europe), is expected to see the steepest regional decline at just over 900 kb/d, to 12.8 mb/d by 2030. Refinery runs are projected to drop nearly 900 kb/d, broadly tracking the fall in demand. Total capacity of 14.5 mb/d is only just above demand for refined products; however, the region's high-cost structure and substantial but uneven product imbalances leave capacity at risk of closure. We expect capacity at risk to rise by 650 kb/d, to 1.6 mb/d.

Arguably, the region's heavy reliance on middle distillate imports have lent support to European diesel and jet fuel cracks that encourage refineries to maximise middle distillate production at the expense of other products. However, the shift in demand away from diesel towards gasoline that has been evident since 2019 has shrunk the net diesel import requirement, particularly for EU nations. The forecast

decline in diesel demand will see this imbalance shrink further in the coming years, albeit the need to import kerosene will increase in parallel.

European refinery runs, demand and net product balances, 2012-2030



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OECD Europe's refining sector remains in structural decline, with major capacity additions since 2011 limited to Repsol's 110 kb/d expansion in 2011 at Cartagena plant in Spain and Socar's 190 kb/d Star refinery in Türkiye in 2019. Against this backdrop, there has been an unbroken trend of rationalisation that has accelerated under mounting economic and policy pressures. Announced closures for the region total of 370 kb/d by 2030, with the potential for substantially more in the latter part of the decade. No expansions or greenfield projects are expected over the forecast period. Beyond investment to meet maintenance and regulatory requirements, or pursue low-carbon fuels and hydrogen integration, the broader stagnation of European refining investment is clear.

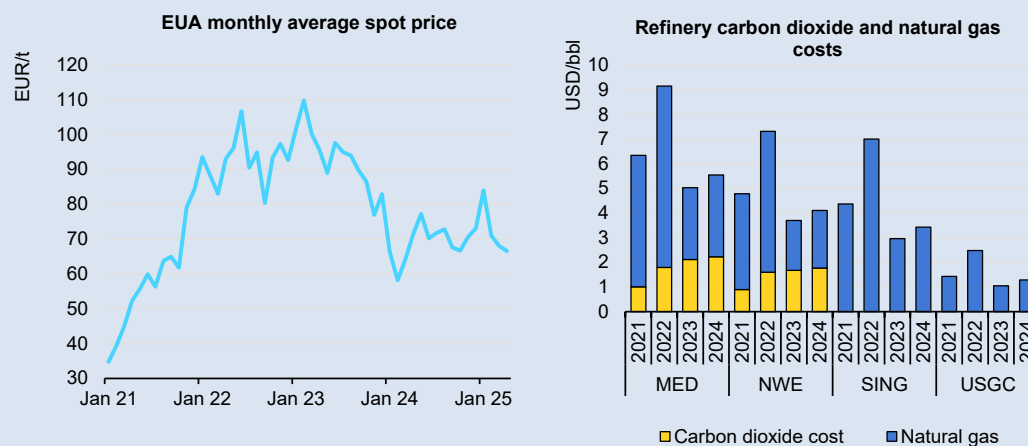
This year marks the region's largest annual capacity reduction since 2012, with over 370 kb/d of confirmed shutdowns. Shell has closed the 150 kb/d Wesseling section of the Rheinland-Palatinate refinery in Germany, BP is set to shut an 80 kb/d unit at its 240 kb/d Gelsenkirchen refinery before year-end and the 140 kb/d Petroineos Grangemouth facility in the United Kingdom was shuttered in April this year. These closures follow a string of earlier site conversions or shutdowns – including TotalEnergies' 100 kb/d Grandpuits and ExxonMobil's 110 kb/d Slagen in Norway in 2021, Eni's 120 kb/d Livorno, in 2024. Several planned biofuel conversions, such as Grandpuits, Wesseling and Venice, are proceeding, but these involve shutting traditional CDU capacity and thus reducing the region's total crude processing footprint, as well as conversion capacity which cuts light product yields. Moreover, planned investments in renewable hydrogen

projects have also stalled in recent months as companies re-evaluate their strategies.

Regulatory impacts on the European refining system

European refiners face a structurally challenging future as carbon compliance costs rise and as European Union's "Fit for 55" policy ambitions intensify. Chief among these is the EU Emissions Trading System (EU-ETS), which has significantly increased operating costs for refineries since the reduction in free emissions allowances, which began in 2013. The price of EU Allowances (EUA) – the permits required for emitting one tonne of CO₂ under the EU-ETS – has increased from approximately EUR 30/t in 2021 to around EUR 70/t today. This surge represents a significant cost burden for European refiners, translating into a direct carbon cost of around USD 1-2/bbl at current market prices. Historically, the refining sector benefitted from free EUA allocations, calibrated to reward the most carbon-efficient installations and penalise higher-emitting facilities, thereby incentivising emissions reductions. However, these free allocations are being progressively phased out. By 2025, refinery allocations will be down 20% relative to 2020 levels, and by 2030 they are expected to decline by over 40%. This erosion of free allowances will increase the effective carbon cost faced by refiners and may accelerate rationalisation and investment shifts across the sector.

European Union carbon pricing and global refinery costs



Note: MED is Mediterranean, NWE is Northwest Europe, SING is Singapore, USGC is United States Gulf Coast. For further details see the IEA refining margins [methodology](#).

Sources: IEA analysis based on prices from Argus Media Group, All rights reserved.

This rising carbon cost is compounded by elevated natural gas prices, which inflate hydrogen production costs used in desulphurisation processes. The combined cost of natural gas and carbon allowances for Mediterranean refiners averaged nearly USD 9/bbl in 2022 following Russia's invasion of Ukraine. While prices have since eased, the episode exposed the sector's vulnerability to fuel and carbon market

volatility, prompting modest but growing investment in green hydrogen and a greater focus on energy efficiency measures.

At the same time, the European Union's inclusion of maritime transport in the EU-ETS as of 2024 generates additional demand for credits which will boost costs for integrated refining and shipping supply chains. Under the new regime, 100% of emissions from intra-EU voyages and 50% from voyages beginning or ending outside the European Union are now within the scope of compliance. This shift raises freight costs, lowers the refinery netback value of exports and increases the delivered cost of crude.

Furthermore, the expansion of Sulphur Emission Control Areas (including the Mediterranean from 2025), combined with IMO 2020, has pushed refiners to optimise yields for marine gasoil (MGO) and VLSFO. This benefits complex, coastal refineries with desulphurisation capacity and strong middle distillate production, while pressuring simpler plants reliant on high sulphur fuel oil (HSFO) output. In parallel, EU measures such as FuelEU Maritime and the inclusion of shipping in the EU-ETS are expected to erode long-term demand for petroleum-based marine fuels, encouraging investment in biofuel blending, green hydrogen and low-carbon refining upgrades.

Crude and product balances

OECD Europe's crude oil balance will remain in a significant deficit through 2030, sustained by declining regional production and despite easing crude runs. Crude throughputs will drop by nearly 500 kb/d, while crude import requirements will narrow by only 200 kb/d to 5.4 mb/d by the end of the forecast period. As Africa's crude balance tightens, European refiners will be required to rely more on Middle East and Latin American crude imports.

OECD Europe's refined product deficit is expected to remain flat, as both product output and demand declines. Falling gasoline exports and increasing kerosene import needs outpace the shrinking diesel requirements as regional demand contracts by 560 kb/d by 2030. The diesel deficit contracts from 930 kb/d in 2024 to 660 kb/d by the end of the decade.

Challenges

A growing number of European refineries are expected to either close or undergo major transformation by 2030. Most facilities will need to pivot toward an increased share of biorefining, or low-carbon fuel production, to remain operational. Without such adaptation, some face the risk of closure. Operators are already planning to scale back crude distillation, while others explore integration of e-fuels and hydrogen. Rising competition from Middle East and African refineries, which has

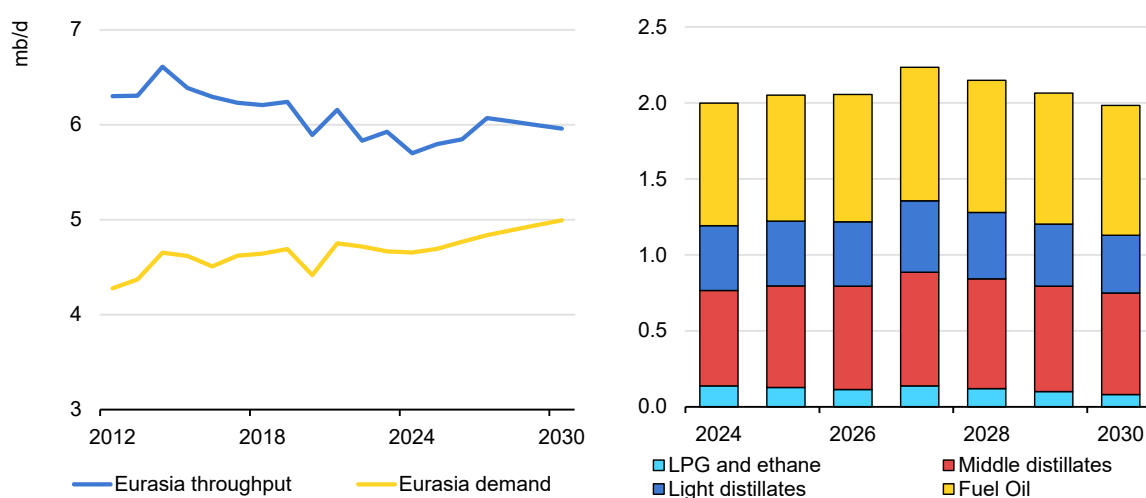
started to limit gasoline exports to West Africa, alongside declining road fuel demand, is accelerating this shift.

Eurasia

Eurasian refining capacity is expected to grow by a modest 180 kb/d over the 2024-30 period, primarily through small-scale expansions and continued modernisation rather than new greenfield developments. This incremental growth reflects both the structural limitations of the regional refining sector and as well as the challenges posed by sanctions, military conflict, ageing assets, and constrained investment flows.

In **Russia**, the most notable recent addition was the 70 kb/d Ust-Luga condensate splitter, which came online in 2024, boosting the site's processing capacity to 220 kb/d. Fortinvest's 60 kb/d expansion at the Afipsky refinery is due online in Q2 2025, while Yug Rusi's 50 kb/d expansion at the Novoshakhtinsk refinery is scheduled for Q1 2026.

Eurasia refinery runs, demand and net product balances, 2012-2030



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Russia has not built a major greenfield refinery in over a decade. In part, this reflects the government's desire for companies to prioritise refinery upgrades. A multi-year modernisation programme has driven a broad wave of investment in secondary units across major sites. These efforts aim to raise conversion rates and boost light product yields. It will also align Russia's refining output slate with domestic and export product specifications. While these investments have only a marginal impact on net distillation capacity, they are critical to sustaining Russia's role as a major refined product exporter.

Elsewhere in Eurasia, refinery projects remain sparse. **Kazakhstan** is advancing a proposed 150 kb/d refinery, though it is still in the feasibility stage and not included in this outlook. Similarly, **Georgia** has broken ground on the Kulevi refinery, but the timing of initial production from the 80 kb/d project is uncertain.

Crude and product balances

The Eurasia region, led by Russia, maintains a substantial net export position in crude oil, though volumes decline slightly over the forecast period. By 2030, total crude exports are projected to fall by nearly 270 kb/d from 2024's level to 7 mb/d. This modest contraction reflects stagnating upstream output amid sanctions on Russia, investment uncertainty, and logistical constraints, even as domestic refinery upgrades modestly lift regional crude runs.

Eurasia will also maintain its refined product surplus at close to 2 mb/d by 2030, supported by gradual improvements in refinery efficiency and higher runs enabled by modernisation efforts. Diesel, naphtha and fuel oil remain the dominant export streams. Altogether, Eurasian crude oil and product exports contract by 300 kb/d over the forecast period but remain robust at 9 mb/d in 2030, reaffirming the region's role as a key supplier of crude and refined products to global markets.

Challenges

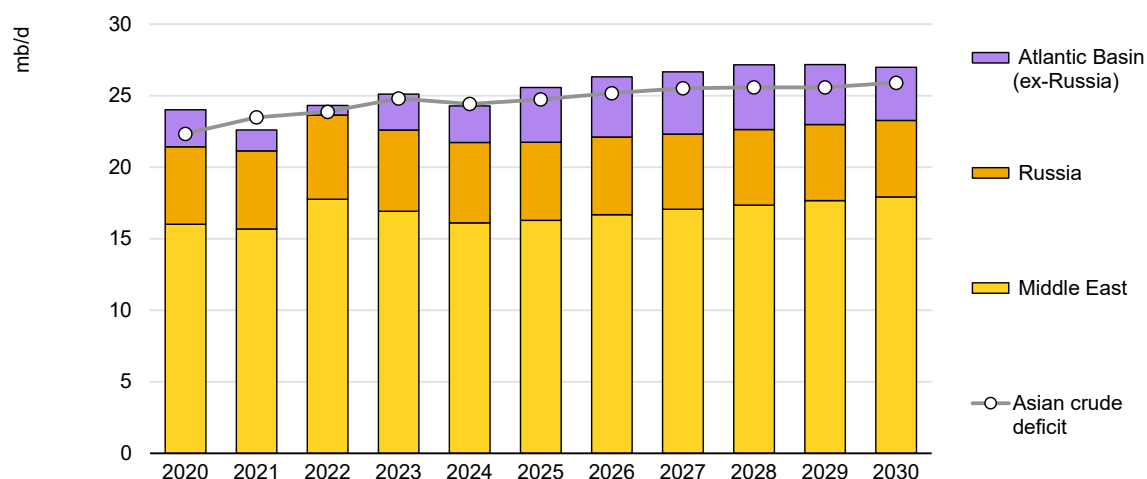
The Eurasian refining sector remains constrained by geopolitical and operational headwinds. Sanctions have slowed equipment procurement and hindered financing for ongoing modernisation, while security risks (including drone strikes on Russian facilities) have temporarily curtailed some operations. The risks that secondary sanctions are imposed on Russian crude and product exports would further complicate the regional outlook. Against the backdrop of steady, upstream output, (in line with this *Report's* assumption that production remains in line with the 31 May 2025 OPEC+ agreement), modestly higher Russian runs lower crude and condensate exports by 270 kb/d by 2030. Although Russian product exports continue to expand, supported by refinery efficiency gains and increased diesel and fuel oil output, sustaining this surplus will require high utilisation rates amid ageing infrastructure and limited foreign investment. Outside Russia, the lack of projects reflect the broader challenge of attracting investment into the sector.

East of Suez

Refining capacity East of Suez is set to expand by a robust 2.7 mb/d by 2030, driven by gross additions exceeding 3.4 mb/d, of which India and China will account for nearly 1 mb/d each. India's growth is driven by large-scale projects, such as Panipat, Barmer and Numaligarh, that are all aimed at meeting rising

domestic fuel demand. China's capacity increase is underpinned by world-scale petrochemical-focused refineries, including Yulong, Huajin and Fujian.

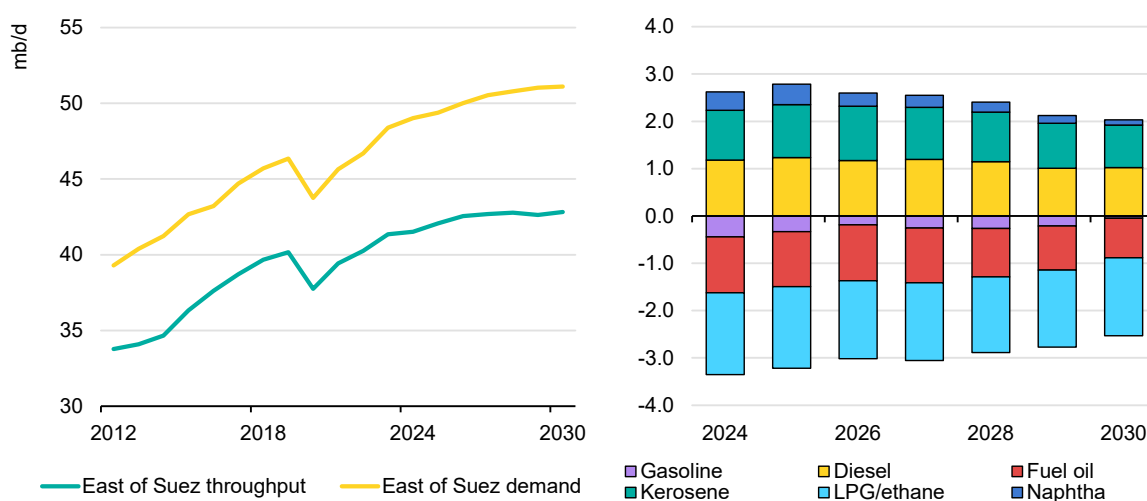
Global crude flows to meet Asian crude deficit, 2020-2030



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However, refining throughput across the region will increase by just 1.3 mb/d – just behind the 1.4 mb/d rise in refined product demand that marginally increases the product supply deficit. More broadly, including non-refined products, total demand growth of 2.1 mb/d is lifted by LPG and ethane growth of 1.2 mb/d while crude oil burn drops by 540 kb/d as Middle East countries push towards gas and renewables in the power sector.

East of Suez refinery runs, demand and net product balances, 2012-2030



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Utilisation rates are expected to fall by 1.6 percentage points to 78.3%, with significant underutilised capacity emerging, particularly in China. Despite adding nearly 1 mb/d of integrated petrochemical-oriented refining capacity, China will see refinery runs increase by just 250 kb/d due to weakening domestic demand for transportation fuels. In the Middle East, capacity growth will moderate after the recent wave of megaprojects, but targeted expansions in Bahrain, Iran and Iraq will still deliver 560 kb/d of net additions. By 2030, the Middle East is expected to contribute an additional 300 kb/d to global refined product supply, further consolidating its role as a key export-oriented refining hub.

Middle East

Following a period of rapid refining capacity expansion from 2020 to 2024 – during which the Middle East added over 2.2 mb/d of net capacity, the region's pace of growth is set to slow. However, strategic investments remain firmly in place through the forecast horizon, with a continued focus on regional fuel security, export competitiveness and condensate processing. By 2030, we expect an additional net 560 kb/d of refining capacity, split almost equally between crude distillation and condensate splitting additions.

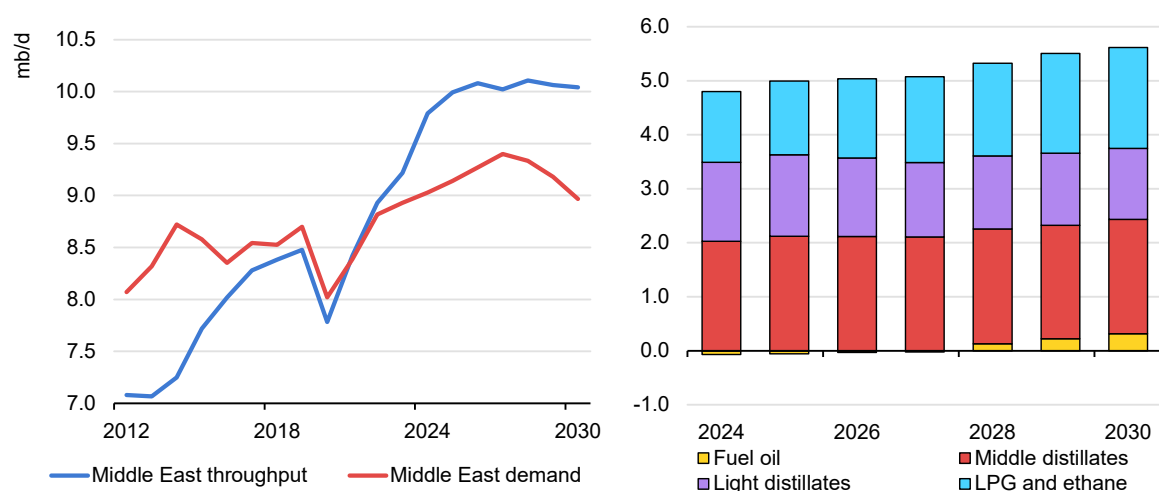
In **Bahrain**, the ongoing Bapco Modernisation Program at the Sitra refinery will add a new 225 kb/d CDU that replaces an older 110 kb/d unit. Associated upgrading units are scheduled to be fully commissioned by Q1 2026, enabling higher throughput and improved product quality, with the site expected to focus more on diesel and jet fuel for export.

Iran's National Iranian Oil Company is pursuing two capacity-enhancing projects. The fourth 120 kb/d train of the Persian Gulf Star condensate splitter is set to be commissioned in Q4 2025. This will be followed by the launch of the 60 kb/d Siraf condensate splitter due online in Q2 2026. Together, these projects will bolster Iran's output of gasoline and other light products, while increasing flexibility in condensate processing.

Oman will see a modest capacity boost from the final 30 kb/d tranche of its Duqm refinery project, due online in H2 2025. Following the inauguration of the export-orientated plant in late 2023, this addition marks the completion of the project and will support further product exports, particularly to Africa and Asia.

Iraq's South Refineries Company is working to bring online the new 100 kb/d Dhi Qar refinery, which we assume will occur in 2028. A further two 70 kb/d expansion projects, at Missan and Diwaniya, are also included before 2030. These projects will reduce gasoline and diesel import dependence in Iraq and meet growing domestic demand, particularly in the transport and power sectors. Several other projects aimed at raising the country's yield of light and middle distillates will also be delivered, lowering fuel oil output, which remains elevated.

Middle East refinery runs, demand and net product balances, 2012-2030



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Crude and product balances

The Middle East will remain a dominant exporter of both crude oil and refined products over the forecast period. Net crude and product exports are projected to rise by 2.7 mb/d to reach 23.7 mb/d by 2030. Growth is driven by steady upstream expansion and increased refining throughput as major projects like Duqm and Jazan ramp up to full utilisation. Net crude oil exports are forecast to grow by 1.8 mb/d by 2030, underpinned by medium and heavy crude production increases in Saudi Arabia, Iraq and the UAE and the 540 kb/d decline in regional use of crude in power generation by 2030. Net product exports will increase by 860 kb/d, led by LPG as rising regional NGLs production boosts exports by 560 kb/d. Declining domestic use of fuel oil lifts exports by 380 kb/d by 2030, while diesel exports gains 120 kb/d. Gasoline and naphtha net exports decline by 150 kb/d, reflecting strong domestic consumption outpacing regional supplies and the shift in refining yields toward diesel and jet fuel.

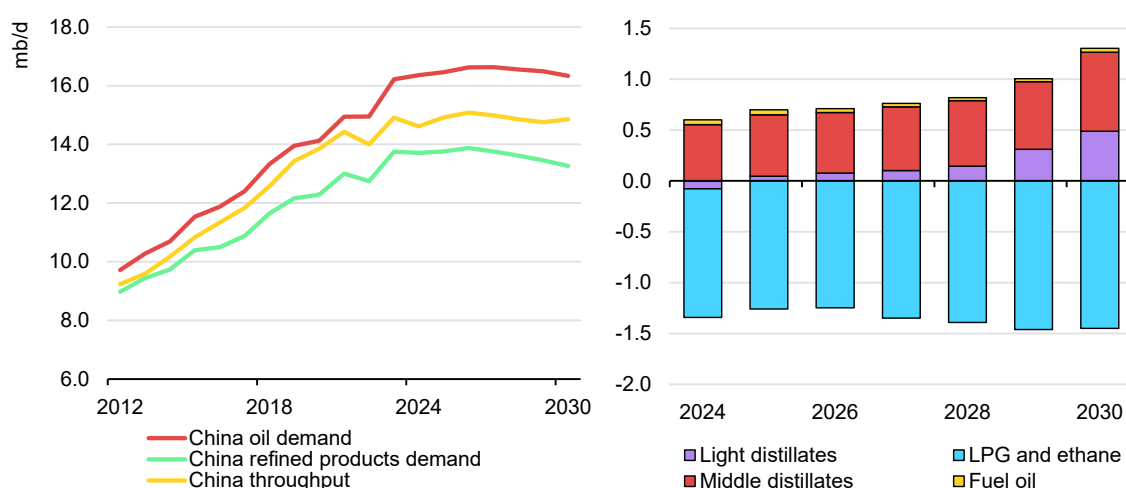
Challenges

While the pace of expansion has slowed compared to the previous five-year period, the Middle East remains a key source of capacity growth, consolidating its role as a refined product export hub while processing heavier and sourer crude grades through upgraded configurations. The region's strong demand growth will weigh on light distillate and jet fuel export volumes, while increasing supplies of medium and heavy crude should start to reward recent investments in upgrading of existing capacity.

China

China's refining landscape will continue to evolve through 2030, albeit under tighter regulatory scrutiny and more deliberate structural adjustments. The country's maturing refining sector will increasingly focus on adapting to domestic energy transition policies, including electrification, efficiency gains and rising non-oil alternatives. Following a surge of large-scale additions over the past decade – which includes Guangdong Petrochemical, Shenghong Petrochemical and Zhejiang – the country's installed capacity has risen to approximately 18.5 mb/d and will see that number swell to 19.5 mb/d by 2030. With Beijing signalling a capacity ceiling near 20 mb/d, future growth will rely heavily on complex, integrated refinery-petrochemical (crude-to-chemicals) hubs, which will be balanced by the closure of small, less efficient operations. Over the forecast horizon to 2030, China is expected to add 1.3 mb/d of new or expanded CDU capacity, while announced closures total approximately 400 kb/d.

Chinese refinery runs, product demand and net product balances, 2012-2030



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The single largest addition will come from the 400 kb/d Yulong Petrochemical complex in Shandong province, which began trial runs in H2 2024. The site's two 200 kb/d CDUs were commissioned in Q1 2025. However, crude processing rates are restricted pending the start-up of the refinery's extensive petrochemical units. This world-scale facility exemplifies China's shift toward vertically-integrated petrochemicals-focused refining systems. Also in 2025, Sinochem's Quanzhou (60 kb/d), CNOOC's Daxie expansion (130 kb/d), and Sinopec's major Zhenhai Refining & Chemical expansion (240 kb/d) are due to start up, boosting capacity by over 400 kb/d in central and eastern China. In 2026, Saudi Aramco and Norinco's 320 kb/d Huajin complex (Panjin II) is also expected to enter operations, forming a key pillar in Aramco's downstream expansion into China. Further out,

Sinopec's Qilu Petchem expansion (70 kb/d) and Sinopec/Fujian's Gulei Phase II project (320 kb/d) are scheduled for 2028 and 2030, respectively.

These follow the model of large-scale, crude-to-chemicals hubs that are central to China's refining strategy. These capacity additions underpin the rapid shift in Chinese refinery yields that is necessary to reinforce the country's pivot to petrochemical feedstock production and away from traditional transportation fuels. From current levels, we forecast that naphtha yields will increase by 3.6%, with a corresponding decline in gasoline yields. Similarly, kerosene yields are boosted by 1.2% at the expense of diesel.

Offsetting these additions, the government continues to push the closure of small, inefficient refineries in Shandong. Three independent "teapots" – Wudi Xinyue, Dongfang Hualong, and Shouguang Luqing – will shut between 2026 and 2028, accounting for 170 kb/d in lost capacity. Other closures include Sinochem's Daqing Zhonglan (50 kb/d) and CNPC's Dalian WEPEC site, which will retire two CDUs totalling 200 kb/d in 2025. These closures, while significant, are dwarfed by the scale and complexity of new capacity additions.

Crude and product balances

China will remain the world's largest net importer of crude through the medium term, rising from 10.5 mb/d in 2024 to 11 mb/d by 2030. This reflects both falling domestic crude output and refining throughput that increases from 14.6 mb/d to 14.9 mb/d over the same period. This marks a slowdown from the long-term trend of rapidly increasing crude imports.

Ethane and LPG imports are forecast to increase from 1.3 mb/d in 2024 to close to 1.5 mb/d by 2030. Naphtha imports will stabilise at close to 400 kb/d, with sharply higher domestic supply at the expense of gasoline production. Nevertheless, the contraction in domestic gasoline demand will boost the potential for exports towards the end of the decade. Despite the assumed shift to jet fuel production that will stabilise jet fuel exports at current levels, a decline in diesel demand of nearly 430 kb/d by 2030 lifts diesel exports from around 200 kb/d in 2024 to 450 kb/d by the end of the decade.

Challenges

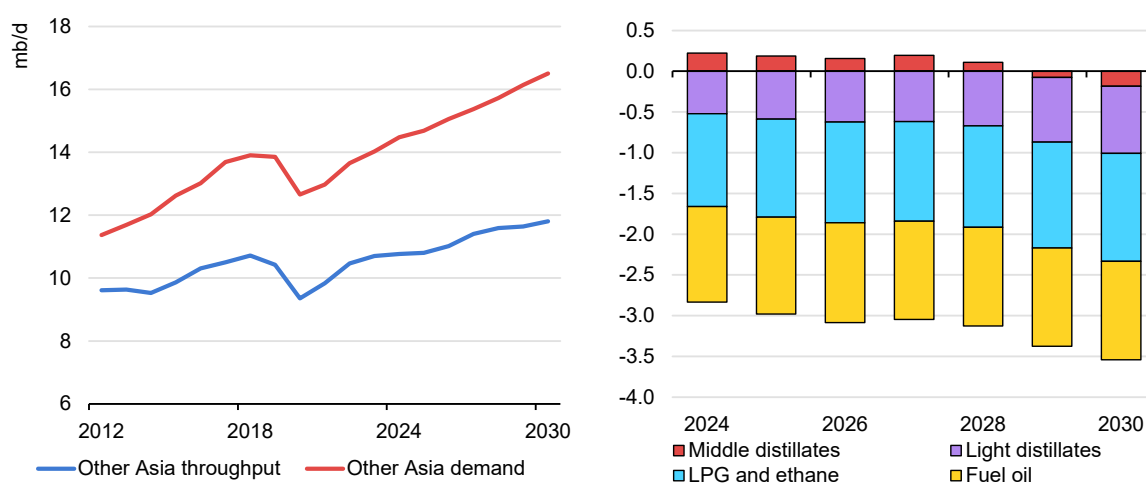
Amid lower domestic transport fuel demand and continued refinery capacity additions, utilisation rates are expected to remain under pressure. In 2024, national refinery utilisation averaged around 79%. We forecast that it will drop further to 76% as transportation fuel demand declines. Historically, Chinese refiners have relied on clean product export quotas of around 40 Mt/yr (880 kb/d) to manage surplus output, with diesel and gasoline exports rising sharply in 2022-23 during periods of favourable margins. We assume these volumes will

persist, as refiners seek to capitalise on rising import requirements in Southeast Asia. By the end of the decade, China's refining capacity could approach the 20 mb/d ceiling, with new additions requiring closures elsewhere. This reflects China's strategic pivot from transportation fuels towards large, integrated crude to chemical oriented complexes and rationalising its fragmented domestic-oriented capacity.

Other Asia

Refining capacity in Other Asia is expected to increase by 1.3 mb/d to reach 14.3 mb/d by 2030, driven almost entirely by large expansion projects in India. The region will see gross additions of 1.4 mb/d, partially offset by 170 kb/d of closures. While capacity growth in the region supports a forecast increase of 1 mb/d in refinery throughput, the region's older and smaller refineries are expected to struggle with increased competition from the Middle East. This is expected to weigh on regional crude runs growth, even though they remain insufficient to fully meet the 1.8 mb/d increase in regional refined product demand.

Other Asia refinery runs, demand and net product balances, 2012-2030



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In contrast to China, where capacity growth is slowing amid policy caps and demand saturation, **India** continues to expand aggressively to meet rising domestic fuel demand and to cement its position as a regional refining and product export hub, accounting for nearly 1 mb/d of new capacity. Much of the new capacity prioritises chemical integration, reducing flexibility to meet diesel and gasoline demand. The country's capacity build-out is dominated by several refinery upgrades. The major expansion at the Panipat refinery will add 200 kb/d, with start-up scheduled for 2026. Elsewhere, capacity increases are expected at Numaligarh (+120 kb/d), Koyali (+80 kb/d), Bina (+60 kb/d). We also include the

180 kb/d increase at Nagapattinam, albeit not until 2030, alongside smaller upgrades at Mumbai, Digboi, and Kochi. The 180 kb/d Barmer refinery – the only greenfield project in the forecast – is scheduled to come online in 2026. The Barauni refinery will see a new 130 kb/d CDU installed to replace an old 70 kb/d CDU. Capacity additions in 2026 alone are expected to reach 680 kb/d, making it the single largest year for Indian refining growth in more than a decade.

Elsewhere in the region, additions are aimed at meeting local demand. **Indonesia** will add 100 kb/d of capacity at Balikpapan in 2026. In **Thailand**, Thai Oil is upgrading its Sriracha facility, replacing older crude units with a new 200 kb/d CDU by 2028, resulting in a net addition of 105 kb/d. **Brunei's** Hengyi Petrochemical refinery is advancing a second-phase 55 kb/d addition at its Pulau Muara Besar site, expected by 2029. **Mongolia** is also proceeding with its 30 kb/d greenfield Sainshand refinery, its first, scheduled for 2029.

Crude and product balances

Crude import requirements in Other Asia are set to rise by 1.4 mb/d to reach 9.9 mb/d by 2030 – making it the second largest import-dependent region globally, after China. India, the key growth driver, continues to pursue the acquisition of cost-advantaged crudes, replacing a swathe of Middle East grades with 1.8 mb/d of Russian crude imports in 2024, even as the Urals discount versus benchmark Dubai crude narrowed. The region's rising crude needs will be met through increased inflows from the Atlantic Basin as well as the Middle East, which benefits from its closer proximity and established contractual ties.

Despite this increase in regional capacity and rising crude imports, Other Asia is projected to face a growing refined product shortfall. Net product import requirements increase by 1 mb/d to 3.5 mb/d over the forecast period. Middle distillates see the largest shift, moving from a 220 kb/d surplus in 2024 to a 180 kb/d deficit by 2030. The regional gasoline deficit widens by around 110 kb/d in tandem with a 200 kb/d deterioration in naphtha balances, as petrochemical feedstock requirements surge. Elsewhere, clean cooking demand supports a 270 kb/d increase in LPG demand that pushes regional import needs above 1.2 mb/d by 2030. Fuel oil import dependence remains high, particularly in Singapore, the world's key marine fuel bunkering hub.

Challenges

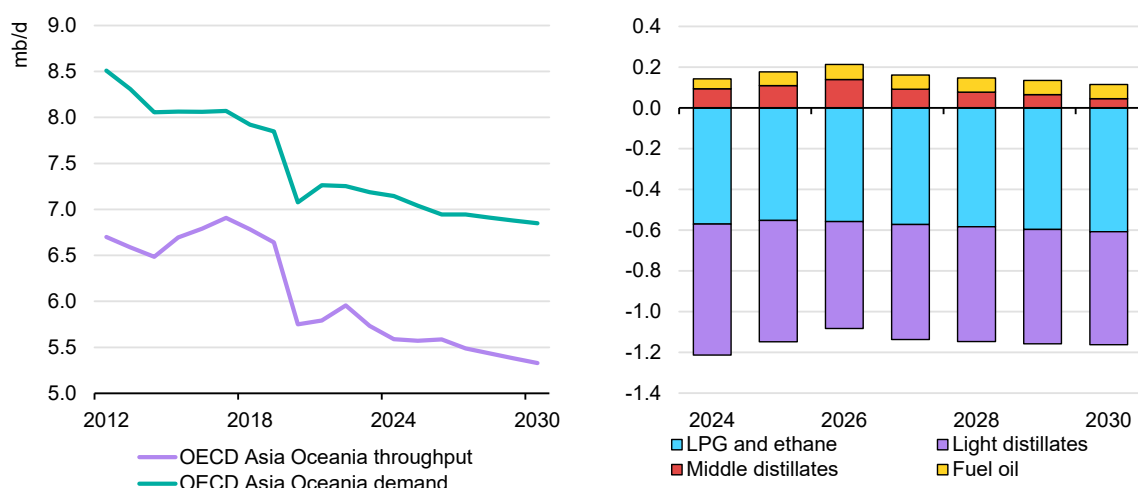
Other Asia became the largest net importer of refined products in 2024, surpassing Africa. Strong demand growth, particularly for transport fuels and petrochemical feedstocks, continues to outpace refining output, driving product import needs up by nearly 1 mb/d by 2030. Despite steady capacity additions, primarily in India,

the region faces persistent structural and rising import needs, with the crude import dependence also rising by 1.4 mb/d.

OECD Asia Oceania

OECD Asia Oceania distillation capacity is forecast to remain unchanged through 2030. This reflects a lack of announcements since the last regional closure of the 120 kb/d Yamaguchi refinery in Japan in 2024. However, the 300 kb/d decline in regional demand for refined products by 2030 will weigh on utilisation rates and we have lowered the crude throughput forecast by 260 kb/d accordingly. Both Korea and Japan are expected to see crude runs fall by a combined 240 kb/d through 2030, despite both countries retaining the need to import petrochemical feedstocks. The start of S-Oil's Shaheen petrochemical expansion at the Onsan plant in Korea will lift demand for feedstocks, even as transportation fuel demand contracts.

OECD Asia Oceania refinery runs, demand and net product balances, 2012-2030



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Crude and Product balances

Crude import requirements in OECD Asia Oceania are set to decline by 200 kb/d through 2030, as upstream output proves more durable than refined product demand and crude throughputs. Refined product import requirements remain stable, at around 650 kb/d, albeit heavily skewed towards LPG and naphtha – key feedstocks for Japan and Korea's petrochemical industries. Declining gasoline and diesel demand modestly lift the need to export surplus product, largely from Japan.

Challenges

Refining capacity in OECD Asia Oceania continues to struggle to compete with more modern and complex capacity elsewhere in Asia. Structural demand erosion, ageing assets and limited reinvestment compound the problem. Japan has paused the phase out of older capacity for now, while New Zealand has exited refining entirely. Although Korea maintains a robust, export-oriented refining sector, the region as a whole is shifting its focus toward petrochemical integration, energy efficiency and alternative fuels. These structural shifts are reinforcing OECD Asia Oceania's reliance on imported refined products.

Natural gas liquids

Global summary

Steady supply growth meets rising East of Suez demand

Growing global natural gas liquids (NGLs)¹ supply plays an outsized role in demand in our medium-term outlook. Rapidly increasing production from natural gas fields and associated gas feeds into key streams for petrochemical feedstocks, such as ethane, liquefied petroleum gas (LPG) and pentanes plus (C5+) or naphtha, as well as for gasoline blending, cooking and heating. Upstream oil projects involving lighter and gassier fields, especially in North America and the Middle East, will boost NGL output by 2.0 mb/d in the 2024-30 forecast, to 15.5 mb/d, accounting for nearly 50% of all global supply gains over the period.

The industry sends the vast majority of these flows to fractionation units – often centralised to cut costs – where they are separated into their components. However, around 10% of NGLs produced are used directly, mainly in refining and generally by blending into crude or bitumen.

Petrochemical producers have invested in new capacity to take advantage of growing availability of low-cost feedstocks, particularly ethane and LPG. Abundant LPG will also allow increased use for clean cooking in developing regions. Competition from these feedstocks will be a major factor in marginalising the role of the refining industry in supplying petrochemical demand to 2030.

Demand for ethane, an essential petrochemical feedstock, grows by 610 kb/d to 5.2 mb/d by 2030. The United States, the world's largest producer and consumer, will see continued production gains feeding both higher domestic demand as well as exports to Asian and European customers. Increased Middle East supply will mostly fuel expanding regional industrial use.

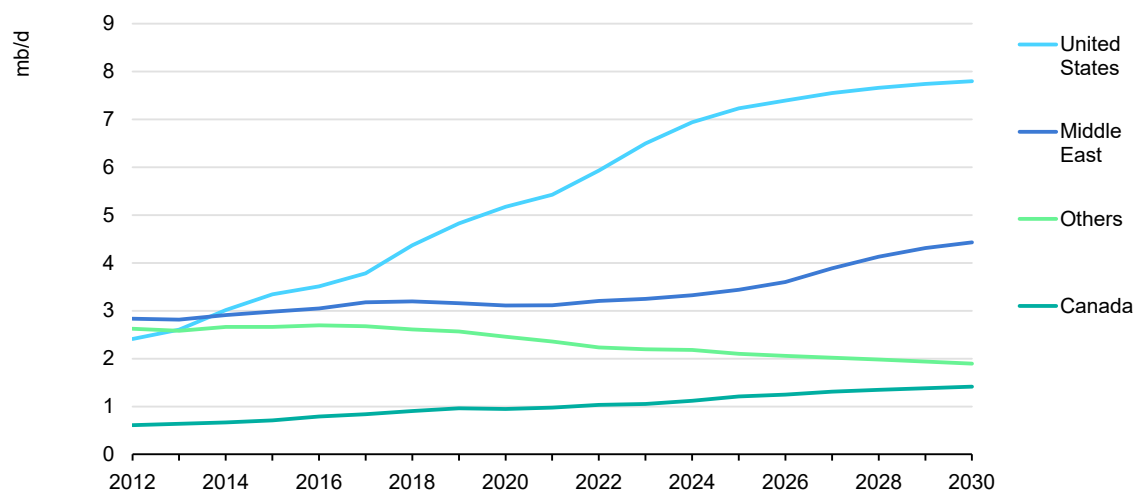
Overall LPG consumption will increase by 1.3 mb/d to 11.8 mb/d from 2024 to 2030, led by stronger demand in petrochemicals but also from rising use in heating and clean cooking. East of Suez demand expands by 820 kb/d to 6.6 mb/d, amounting to over 65% of the global gains, led by China and India. Atlantic Basin demand is driven by Africa and the United States. Canada, the United States and the Middle East will continue to dominate global LPG exports through to 2030.

¹ NGLs in this section does not include any condensate, including for OPEC+ countries

Supply growth driven by North America and Middle East

NGLs production has steadily ramped up over the past decade, reflecting burgeoning supply from unconventional reserves as well as upstream projects with lighter and gassier crude fields and wetter gas fields, especially in North America. The fractionation of NGLs at or near the wellhead yields ethane, propane, butanes and heavier fractions such as pentanes plus (C5+).

Global NGLs output, 2012-2030



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From 2014 to 2024, global NGLs production grew by 4.3 mb/d to 13.6 mb/d. NGLs will rise by a further 2.0 mb/d to 15.5 mb/d in 2030, with average annual growth slowing to 2.3% over the forecast period, from 3.9% during the previous decade.

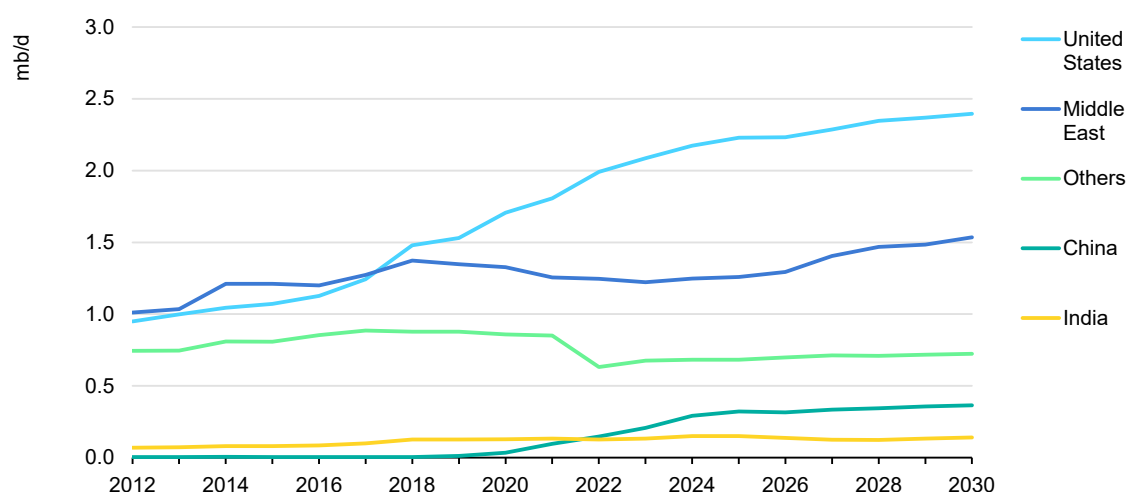
The United States, Saudi Arabia and Canada dominate global NGLs supply, with the three countries accounting for 84% of production today and rising to 88% by 2030. The United States alone held an overwhelming 51% market share, with production set to increase from 6.9 mb/d in 2024 to 7.8 mb/d in 2030 (+860 kb/d). Canadian output will rise by 300 kb/d, from 1.1 mb/d to 1.4 mb/d over the period. Despite increased supply in both countries, North America's share of global production remains unchanged through 2030 at around 60%. The Middle East will boost its share from 25% to 29% over the same period as output expands by 1.1 mb/d to 4.4 mb/d. Saudi Arabian production will rise to 2 mb/d by 2030, from 1.4 mb/d in 2024, supported by investments in natural gas and unconventional fields. Additional supply will also come from the buildout of natural gas and LNG projects in the UAE (+210 kb/d) and Qatar (+190 kb/d). By contrast, falling production in Asia and Europe (-110 kb/d, each) will cut their modest shares by 1%, to 3% in Asia and to 1% in Europe. Demand for ethane and LPG in the two regions will rise through 2030, however, increasing their import requirements.

Ethane markets

Global ethane markets maintain their US dependence on the United States

The global ethane market has a long-established supply-demand structure, with demand generally restrained by limited supply outside the United States and the Middle East. The United States dominates global output with an ample surplus of extractable volumes from low-cost and steadily growing natural gas streams. This makes it both the world's leading exporter and consumer of ethane, far exceeding other large consumers, including the Middle East, Canada, China and India. Middle Eastern countries are self-sufficient, using ethane from their own gas production. But China, Europe and India, and to a lesser extent Canada, are all dependent on imports of US ethane. Little will change in this configuration through 2030, though rising Middle East gas production will provide additional ethane to meet growth in local demand.

Global ethane demand, 2012-2030



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Total ethane demand expands by 610 kb/d from 2024 to 2030, reaching 5.2 mb/d. The Middle East will lead demand growth, with project developments largely mirroring forecast increases in natural gas production. Regional demand increases by 290 kb/d to 1.5 mb/d in 2024-30, led by Saudi Arabia (+160 kb/d to 710 kb/d), Qatar (+100 kb/d to 250 kb/d) and the UAE (+40 kb/d to 230 kb/d). European demand will be boosted slightly by a new ethane cracker in Belgium towards the end of the forecast period, while use in Russia and India will remain largely unchanged. Chinese demand rises more slowly after a strong period of growth that lifted uptake to 290 kb/d in 2024. It will gain a further 75 kb/d over the

forecast period, to reach 365 kb/d in 2030. In North America, US demand will increase by 220 kb/d to 2.4 mb/d by 2030, while Canada stagnates at 240 kb/d in the absence of new projects.

US ethane production is forecast to expand by 380 kb/d over the forecast, to reach 3.1 mb/d by 2030, boosting volumes available for export from 520 kb/d to 680 kb/d. Recently announced US export licencing obligations to China could impact this outlook. Growth in US exports offsets the deepening deficits in China and Europe by 2030. Rising production in Canada narrows the country's net shortfall by 70 kb/d to 25 kb/d in 2030.

Ethane storage capacity reached 162 mb in the United States in 2024, according to *Argus Media*, covering roughly 60 days of supply for this net exporter. Canadian storage of 26 mb covers about 108 days of demand and China's 11.5 mb (increasing to 12.8 mb in 2025) covers 39 days of current demand.

LPG markets

Abundant global LPG supply supports demand growth

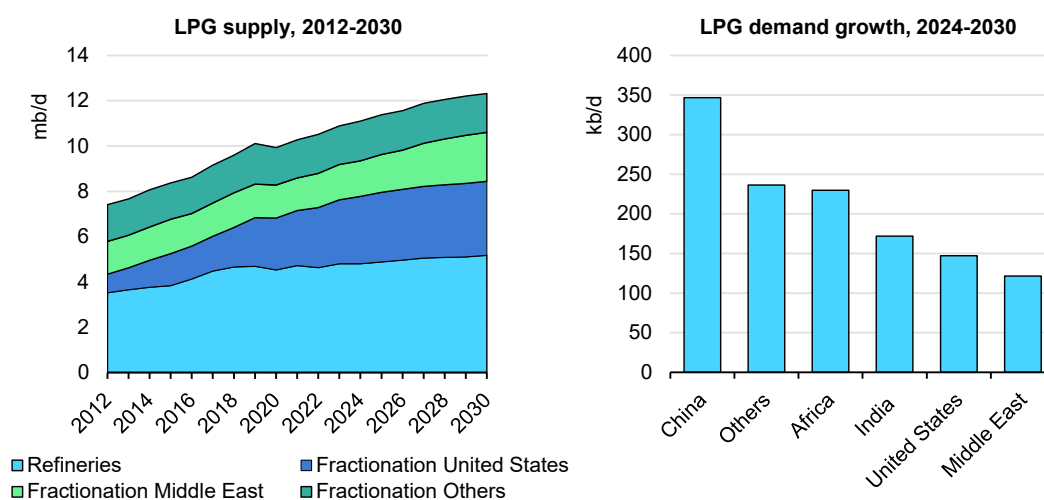
LPG supply remains abundant, weighing on prices relative to competing refined oil products and continuing to support strong demand growth as it has in recent years. Global LPG demand will rise by 1.3 mb/d to 11.8 mb/d from 2024 to 2030. Today, around 43% of LPG comes from refineries and the rest from NGL fractionation. LPGs include propane, normal butane and isobutane. LPG is used for cooking and heating (47%), petrochemical feedstocks (30%), industry (9%), transportation fuels (6%) and in the energy sector (6%). Expanding use in petrochemicals, cooking and heating drive demand growth through 2030.

Most incremental LPG supply comes from fractionation of NGLs, which is forecast to rise 850 kb/d to 7.1 mb/d by 2030. Gains are dominated by the Middle East (+600 kb/d to 2.2 mb/d) and North America (+400 kb/d to 4.1 mb/d). Saudi Arabia alone raises fractionation output by +390 kb/d to 1.0 mb/d. Refineries represent only 30% of incremental supply over 2024-30, increasing by 370 kb/d to 5.2 mb/d. Almost all this additional refinery supply arises from new capacity in China and India, while European output falls due to plant closures.

Markets East of Suez account for two-thirds of overall demand growth, gaining 820 kb/d over the outlook period to reach 6.6 mb/d. Of this, China's consumption increases by 350 kb/d to 2.7 mb/d, boosting its share of world demand from 22% to 23%. India follows with a 170 kb/d rise to 1.0 mb/d, underpinned by new petrochemical capacity and growing residential use, particularly in clean cooking.

The Middle East's appetite for LPG will increase 120 kb/d to 790 kb/d by 2030, mainly led by Saudi Arabia (+64 kb/d). In Africa, rapidly expanding use of LPG for heating and cooking boosts uptake by 230 kb/d to 860 kb/d. Demand in North Africa and West Africa each expand by 90 kb/d, to 540 kb/d and 220 kb/d, respectively, while use doubles in the rest of Africa to 100 kb/d. North American demand increases by 150 kb/d to 1.9 mb/d, almost entirely in the United States. European LPG consumption stagnates at around 1.1 mb/d through 2030 while Eurasian use will rise 11% to 920 kb/d by the end of the decade.

Global LPG supply and demand



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Stable demand growth East of Suez, especially in Asia, continues to attract LPG supplies from major exporting countries. Available LPG for export from the United States and Canada reaches 2.6 mb/d, up 310 kb/d in the 2024-30 period, while Middle East exports grow 490 kb/d to just over 1.7 mb/d. Combined, these exporters account for 92% of all global exports, rising to 95% in 2030. The massive export flows partially help to offset widening deficits in Asia (from -2.6 mb/d in 2024 to -2.9 mb/d in 2030), Africa (-340 kb/d to -570 kb/d) and Europe (-480 kb/d to -600 kb/d). China's supply deficit increases by 110 kb/d to -1.1 mb/d, while the ASEAN deficit widens by 120 kb/d to -590 kb/d.

LPG storage capacity worldwide reached 709 mb in 2024, according to data from *Argus Media*, covering almost 70 days of global demand. North America accounts for 50% of the world's storage with capacity of 354 mb, of which 41% in the United States (295 mb), 6% (44 mb) in Canada and 2% in Mexico (15 mb). European storage amounts to 49 mb, equal to 45 days of demand. Capacity in OECD Asia Oceania was 71 mb in 2024, or 92 days of demand, with Japan accounting for over two-thirds of the region capacity at 48 mb, representing 121 days of demand while Korea holds 23%, or 16 mb, covering 49 days of demand. The Middle East

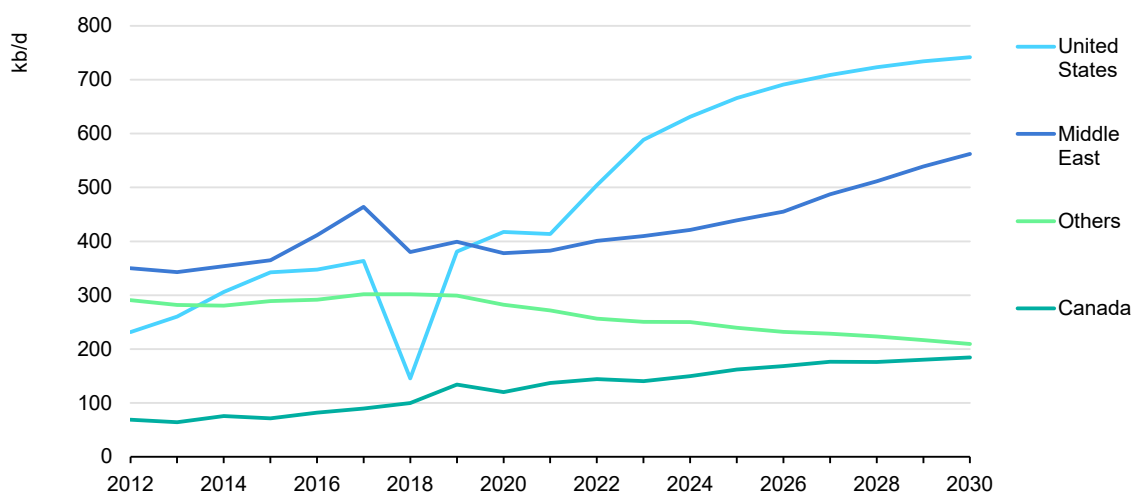
accounts for 7% of global capacity, or 52 mb, covering 77 days of regional demand. China, the second largest user of LPG, holds 11% of world storage capacity, or 78 mb, equal to 33 days of demand. An additional 4.8 mb of capacity will be installed in China in the next few years, according to *Argus Media*. India will also raise capacity by 1.4 mb from 11 mb while Brazil will add 0.8 mb. European capacity will rise marginally (+0.4 mb).

C5+ markets

Increased NGL fractionation boosts Naphtha/C5+ supply

Global naphtha demand is met by supply from both refining and NGL fractionation that which produces a mix of molecules, referred to as C5+ and sometimes as natural gasoline, that are heavier than butane and are similar to naphtha. Most naphtha production comes from refineries (rising from 8.1 mb/d to 8.9 mb/d in 2024-30). Supply of C5+ is forecast to rise 250 kb/d to 1.7 mb/d by the end of the forecast. The output goes to refining and blending requirements but also directly for blending into the crude supply. For example, last year in the United States 45% of C5+ was transferred to the crude stream, around 30% went directly to refiners and blenders, while the rest was exported elsewhere as naphtha.

Global C5+ from fractionation, 2012-2030



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Most of the world's C5+ is produced in the United States and Canada, where supply rises from 630 kb/d to 740 kb/d and from 150 kb/d to 180 kb/d, respectively, over 2024-30, for a combined gain of 140 kb/d. The Middle East accounts for another 140 kb/d of the global increase, reaching 560 kb/d by 2030.

Tables

Table 1
WORLD OIL SUPPLY AND DEMAND
(million barrels per day)

	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
OECD DEMAND										
Americas	24.0	24.7	25.0	24.9	25.0	24.9	24.9	24.7	24.5	24.4
Europe	13.1	13.6	13.5	13.5	13.5	13.4	13.2	13.0	12.9	12.7
Asia Oceania	7.3	7.3	7.2	7.2	7.1	7.0	7.0	7.0	6.9	6.9
Total OECD	44.4	45.6	45.7	45.7	45.6	45.3	45.1	44.7	44.4	44.0
NON-OECD DEMAND										
Eurasia	4.7	4.7	4.7	4.7	4.7	4.8	4.9	4.9	5.0	5.0
Europe	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.9	0.9
China	15.2	15.2	16.5	16.6	16.7	16.9	16.9	16.9	16.8	16.7
Other Asia	13.3	14.1	14.5	15.0	15.3	15.7	16.1	16.4	16.9	17.3
Latin America	5.9	6.2	6.3	6.4	6.5	6.6	6.7	6.8	7.0	7.1
Middle East	8.6	9.1	9.2	9.2	9.4	9.5	9.6	9.6	9.4	9.2
Africa	4.4	4.5	4.6	4.6	4.8	4.9	5.0	5.2	5.3	5.4
Total Non-OECD	52.9	54.4	56.5	57.4	58.2	59.2	60.1	60.7	61.2	61.5
Total Demand¹	97.4	100.0	102.2	103.0	103.8	104.5	105.1	105.4	105.6	105.5
OECD SUPPLY										
Americas	24.4	25.8	27.5	28.3	28.9	29.0	29.3	29.3	29.2	29.2
Europe	3.4	3.2	3.2	3.2	3.3	3.3	3.1	3.1	2.9	2.7
Asia Oceania	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Total OECD²	28.3	29.5	31.1	31.9	32.6	32.8	32.8	32.7	32.5	32.2
NON-OECD SUPPLY										
Eurasia	13.8	13.9	13.8	13.5	13.6	13.7	13.7	13.6	13.5	13.5
Europe	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
China	4.1	4.2	4.3	4.3	4.4	4.4	4.4	4.3	4.3	4.2
Other Asia	2.9	2.7	2.7	2.6	2.6	2.5	2.5	2.5	2.5	2.4
Latin America	5.3	5.7	6.2	6.4	6.8	7.2	7.3	7.7	7.8	7.6
Middle East	3.1	3.2	3.1	3.1	3.1	3.2	3.3	3.5	3.6	3.7
Africa	2.5	2.5	2.5	2.5	2.5	2.5	2.6	2.6	2.5	2.4
Total Non-OECD²	31.7	32.3	32.7	32.6	33.1	33.5	33.8	34.3	34.3	33.9
Processing gains ³	2.2	2.3	2.4	2.4	2.4	2.5	2.5	2.5	2.5	2.5
Global Biofuels	2.8	2.9	3.1	3.4	3.5	3.7	3.8	3.9	4.0	4.1
Total Non-OPEC	65.0	66.9	69.4	70.3	71.6	72.4	72.8	73.4	73.2	72.7
OPEC⁴										
Crude	25.4	27.7	27.4	27.2						
NGLs	5.3	5.5	5.5	5.5	5.7	5.9	6.3	6.5	6.7	6.9
Total OPEC	30.7	33.1	33.0	32.8						
Total Supply	95.7	100.0	102.3	103.1						
Memo items:										
Call on OPEC crude + Stock ch. ⁵	27.1	27.6	27.3	27.2	26.5	26.2	26.1	25.5	25.6	26.0

¹ Measured as deliveries from refineries and primary stocks, comprises inland deliveries, international marine bunkers, refinery fuel, crude for direct burning, oil from non-conventional sources and other sources of supply. Includes biofuels.

² Comprises crude oil, condensates, NGLs, oil from non-conventional sources and other sources of supply.

³ Net volumetric gains and losses in the refining process and marine transportation losses.

⁴ OPEC includes current members throughout the time series.

⁵ Total demand minus total non-OPEC supply and OPEC NGLs.

For the purpose of this and the following tables :

- OECD comprises of Australia, Austria, Belgium, Canada, Chile, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Israel, Italy, Japan, Korea, Latvia, Lithuania, Luxembourg, Mexico, Netherlands, Norway, New Zealand, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland, Republic of Türkiye, UK, US.

- OPEC comprises of Algeria, Congo, Equatorial Guinea, Gabon, Iran, Iraq, Kuwait, Libya, Nigeria, Saudi Arabia, UAE, Venezuela.

Table 1a
WORLD OIL SUPPLY AND DEMAND: CHANGES FROM OIL 2024
(million barrels per day)

	2022	2023	2024	2025	2026	2027	2028	2029	2030
OECD DEMAND									
Americas	0.0	0.0	0.0	0.0	0.2	0.4	0.6	0.8	1.0
Europe	0.0	0.0	0.2	0.3	0.3	0.3	0.3	0.4	0.4
Asia Oceania	0.0	0.0	-0.1	-0.1	-0.2	-0.2	-0.1	-0.1	-0.1
Total OECD	0.0	0.0	0.2	0.2	0.3	0.6	0.8	1.0	1.2
NON-OECD DEMAND									
Eurasia	-0.2	-0.2	-0.2	-0.2	-0.2	-0.3	-0.3	-0.3	-0.3
Europe	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
China	0.1	0.1	-0.5	-0.8	-0.8	-0.9	-1.1	-1.2	-1.4
Other Asia	-0.1	-0.1	0.1	0.0	-0.1	-0.1	-0.1	-0.1	-0.1
Latin America	-0.1	-0.1	-0.1	-0.1	-0.1	0.0	0.0	0.1	0.1
Middle East	0.2	0.2	0.2	0.1	0.2	0.3	0.3	0.3	0.2
Africa	0.1	0.1	0.2	0.3	0.2	0.2	0.3	0.3	0.3
Total Non-OECD	-0.1	0.0	-0.4	-0.7	-0.8	-0.7	-0.9	-1.0	-1.2
Total Demand	-0.1	0.0	-0.2	-0.5	-0.5	-0.2	-0.1	0.0	0.1
OECD SUPPLY									
Americas	0.1	0.1	0.2	0.1	-0.1	0.0	-0.1	-0.2	-0.3
Europe	0.0	0.0	0.0	-0.1	0.1	0.1	0.0	0.0	-0.1
Asia Oceania	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total OECD	0.1	0.1	0.1	0.0	0.1	0.1	-0.1	-0.2	-0.4
NON-OECD SUPPLY									
Eurasia	0.0	0.0	0.0	-0.1	-0.2	-0.1	-0.2	-0.3	-0.3
Europe	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
China	0.0	0.0	-0.1	-0.1	0.0	0.1	0.1	0.2	0.2
Other Asia	0.0	0.0	0.0	0.1	0.1	0.2	0.3	0.3	0.4
Latin America	0.0	0.0	-0.1	0.0	-0.3	-0.4	-0.4	-0.4	-0.3
Middle East	0.0	0.0	0.0	0.0	0.0	-0.1	-0.1	0.0	0.1
Africa	0.0	0.0	0.0	-0.2	-0.2	-0.2	-0.1	-0.2	-0.3
Total Non-OECD	0.0	0.0	-0.3	-0.4	-0.6	-0.5	-0.4	-0.4	-0.2
Processing Gains	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Global Biofuels	0.0	0.0	0.1	0.1	0.2	0.3	0.3	0.4	0.3
Total Non-OPEC	0.1	0.1	0.0	-0.2	-0.4	-0.2	-0.1	-0.2	-0.2
OPEC									
Crude	-0.5	-0.3							
NGLs	0.1	0.0	0.0	0.0	0.0	0.1	0.2	0.2	0.2
Total OPEC	-0.4	-0.3							
Total Supply	-0.3	-0.2							
Memo items:									
Call on OPEC crude + Stock ch.	-0.3	-0.1	-0.2	-0.2	-0.1	-0.1	-0.2	0.0	0.1

Table 1b
WORLD OIL SUPPLY AND DEMAND - WEO Regions
(million barrels per day)

	2022	2023	2024	2025	2026	2027	2028	2029	2030
DEMAND									
North America	24.3	24.6	24.5	24.6	24.5	24.4	24.3	24.1	24.0
Central and South America	6.6	6.7	6.8	6.9	7.0	7.1	7.2	7.4	7.4
Europe	14.3	14.2	14.3	14.3	14.2	14.0	13.9	13.7	13.5
Africa	4.5	4.6	4.6	4.8	4.9	5.0	5.2	5.3	5.4
Middle East	9.1	9.2	9.2	9.4	9.5	9.6	9.6	9.4	9.2
Eurasia	4.7	4.7	4.7	4.7	4.8	4.9	4.9	5.0	5.0
Asia Pacific	36.6	38.2	38.8	39.1	39.6	40.0	40.3	40.7	40.9
Total Demand¹	100.0	102.2	103.0	103.8	104.5	105.1	105.4	105.6	105.5
NON-OPEC SUPPLY									
North America	25.8	27.4	28.3	28.9	29.0	29.3	29.3	29.2	29.2
Central and South America	5.7	6.2	6.4	6.8	7.2	7.3	7.7	7.8	7.6
Europe	3.3	3.3	3.3	3.4	3.4	3.2	3.1	3.0	2.8
Africa	2.5	2.5	2.5	2.5	2.5	2.6	2.6	2.5	2.4
Middle East	3.2	3.1	3.1	3.1	3.2	3.3	3.5	3.6	3.7
Eurasia	13.9	13.8	13.5	13.6	13.7	13.7	13.6	13.5	13.5
Asia Pacific	7.4	7.4	7.4	7.4	7.4	7.3	7.2	7.1	7.0
Total Non-OPEC	61.7	63.9	64.5	65.7	66.3	66.6	67.0	66.8	66.1
Processing gains ³	2.3	2.4	2.4	2.4	2.5	2.5	2.5	2.5	2.5
Global Biofuels	2.9	3.1	3.4	3.5	3.7	3.8	3.9	4.0	4.1
Total Non-OPEC Supply	66.9	69.4	70.3	71.6	72.4	72.8	73.4	73.2	72.7
OPEC⁴									
Crude	27.7	27.4	27.2						
NGLs	5.5	5.5	5.5	5.7	5.9	6.3	6.5	6.7	6.9
Total OPEC	33.1	33.0	32.8						
Total Supply	100.0	102.3	103.1						
Memo items:									
Call on OPEC crude + Stock ch. ⁵	27.6	27.3	27.2	26.5	26.2	26.1	25.5	25.6	26.0

¹ Measured as deliveries from refineries and primary stocks, comprises inland deliveries, international marine bunkers, refinery fuel, crude for direct burning, oil from non-conventional sources and other sources of supply. Includes biofuels.

² Comprises crude oil, condensates, NGLs, oil from non-conventional sources and other sources of supply.

³ Net volumetric gains and losses in the refining process and marine transportation losses.

⁴ OPEC includes current members throughout the time series.

⁵ Total demand minus total non-OPEC supply and OPEC NGLs.

Table 2
SUMMARY OF GLOBAL OIL DEMAND

	2023	2024	2025	2026	2027	2028	2029	2030
Demand (mb/d)								
Americas	24.98	24.94	24.97	24.93	24.86	24.73	24.55	24.36
Europe	13.46	13.51	13.50	13.40	13.20	13.02	12.88	12.68
Asia Oceania	7.24	7.21	7.10	7.00	7.01	6.98	6.95	6.92
Total OECD	45.68	45.67	45.57	45.33	45.07	44.73	44.37	43.95
Asia	30.95	31.63	32.01	32.58	33.01	33.34	33.70	33.96
Middle East	9.16	9.25	9.37	9.50	9.63	9.57	9.41	9.20
Americas	6.34	6.43	6.53	6.62	6.73	6.84	6.95	7.06
Eurasia	4.68	4.67	4.71	4.79	4.86	4.92	4.97	5.03
Africa	4.58	4.59	4.76	4.85	5.01	5.15	5.30	5.45
Europe	0.78	0.80	0.81	0.83	0.84	0.85	0.85	0.86
Total Non-OECD	56.50	57.37	58.19	59.17	60.08	60.66	61.19	61.54
World	102.19	103.04	103.76	104.50	105.15	105.39	105.57	105.50
of which:								
United States ¹	20.28	20.42	20.47	20.45	20.40	20.31	20.15	20.01
Europe 5 ²	7.52	7.57	7.51	7.42	7.28	7.17	7.07	6.93
China	16.48	16.63	16.73	16.89	16.95	16.89	16.81	16.66
Japan	3.29	3.14	3.05	3.01	2.97	2.94	2.91	2.89
India	5.45	5.64	5.75	5.92	6.08	6.25	6.45	6.66
Russia	3.54	3.49	3.50	3.55	3.58	3.59	3.60	3.60
Brazil	3.23	3.33	3.40	3.45	3.49	3.53	3.57	3.60
Saudi Arabia	3.64	3.66	3.67	3.69	3.62	3.49	3.29	3.05
Canada	2.45	2.38	2.39	2.36	2.34	2.31	2.29	2.26
Korea	2.45	2.54	2.52	2.47	2.51	2.51	2.51	2.51
Mexico	1.74	1.74	1.71	1.71	1.71	1.71	1.70	1.70
Iran	1.89	1.92	1.97	1.98	2.01	2.03	2.05	2.07
Total	71.95	72.48	72.67	72.89	72.93	72.73	72.41	71.93
% of World	70.4%	70.3%	70.0%	69.7%	69.4%	69.0%	68.6%	68.2%
Annual Change (% per annum)								
Americas	1.0	-0.2	0.1	-0.2	-0.3	-0.5	-0.8	-0.8
Europe	-0.7	0.4	-0.1	-0.8	-1.5	-1.4	-1.1	-1.6
Asia Oceania	-0.9	-0.5	-1.5	-1.4	0.0	-0.5	-0.4	-0.4
Total OECD	0.2	0.0	-0.2	-0.5	-0.6	-0.7	-0.8	-0.9
Asia	5.8	2.2	1.2	1.8	1.3	1.0	1.1	0.8
Middle East	1.1	1.0	1.3	1.4	1.4	-0.7	-1.6	-2.3
Americas	2.3	1.3	1.6	1.4	1.7	1.7	1.6	1.5
Eurasia	-0.7	-0.2	0.8	1.6	1.5	1.1	1.1	1.1
Africa	2.8	0.2	3.7	2.0	3.4	2.8	2.8	2.8
Europe	3.0	2.7	1.1	2.2	1.0	1.0	0.9	0.4
Total Non-OECD	3.8	1.5	1.4	1.7	1.5	1.0	0.9	0.6
World	2.2	0.8	0.7	0.7	0.6	0.2	0.2	-0.1
Annual Change (mb/d)								
Americas	0.26	-0.04	0.03	-0.04	-0.07	-0.12	-0.19	-0.18
Europe	-0.10	0.06	-0.01	-0.11	-0.19	-0.18	-0.14	-0.20
Asia Oceania	-0.07	-0.03	-0.11	-0.10	0.00	-0.03	-0.03	-0.03
Total OECD	0.10	-0.02	-0.09	-0.24	-0.26	-0.33	-0.36	-0.42
Asia	1.71	0.68	0.38	0.58	0.42	0.33	0.37	0.25
Middle East	0.10	0.09	0.12	0.13	0.13	-0.07	-0.15	-0.21
Americas	0.14	0.08	0.10	0.09	0.11	0.11	0.11	0.10
Eurasia	-0.03	-0.01	0.04	0.08	0.07	0.06	0.05	0.05
Africa	0.12	0.01	0.17	0.09	0.16	0.14	0.15	0.15
Europe	0.02	0.02	0.01	0.02	0.01	0.01	0.01	0.00
Total Non-OECD	2.06	0.87	0.82	0.98	0.91	0.58	0.53	0.35
World	2.16	0.85	0.72	0.74	0.65	0.25	0.17	-0.07

¹ US figures exclude US territories.

² France, Germany, Italy, Spain and UK.

Table 3
WORLD OIL PRODUCTION
(million barrels per day)

	2022	2023	2024	2025	2026	2027	2028	2029	2030
OPEC									
Crude Oil									
Saudi Arabia	10.33	9.57	9.09						
Iran	2.55	2.99	3.34						
Iraq	4.45	4.27	4.31						
UAE	3.30	3.25	3.23						
Kuwait	2.71	2.66	2.55						
Nigeria	1.15	1.24	1.33						
Libya	0.99	1.16	1.07						
Algeria	1.01	0.97	0.91						
Congo	0.26	0.27	0.24						
Gabon	0.19	0.21	0.23						
Equatorial Guinea	0.08	0.06	0.06						
Venezuela	0.64	0.77	0.88						
Total Crude Oil	27.66	27.43	27.24						
<i>of which Neutral Zone¹</i>	0.30	0.37	0.43						
Total NGLs²	5.46	5.53	5.55	5.67	5.91	6.25	6.50	6.70	6.87
Total OPEC³	33.12	32.96	32.79						
NON-OPEC⁴									
OECD									
Americas	25.78	27.46	28.30	28.89	29.03	29.27	29.27	29.24	29.18
United States	18.01	19.52	20.23	20.74	20.90	21.07	21.07	21.08	21.11
Mexico	2.01	2.10	1.97	1.84	1.74	1.60	1.47	1.40	1.29
Canada	5.76	5.83	6.09	6.30	6.38	6.59	6.73	6.75	6.77
Chile	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Europe	3.19	3.23	3.16	3.26	3.32	3.14	3.06	2.88	2.70
UK	0.84	0.73	0.70	0.71	0.72	0.68	0.66	0.64	0.60
Norway	1.91	2.02	2.00	2.03	2.06	1.92	1.88	1.74	1.61
Others	0.45	0.47	0.45	0.53	0.54	0.53	0.51	0.50	0.48
Asia Oceania	0.48	0.46	0.45	0.43	0.41	0.40	0.38	0.37	0.36
Australia	0.41	0.38	0.37	0.36	0.34	0.32	0.32	0.30	0.30
Others	0.07	0.07	0.08	0.07	0.07	0.07	0.07	0.07	0.07
Total OECD	29.46	31.14	31.90	32.58	32.76	32.80	32.72	32.50	32.23
NON-OECD									
Eurasia	13.91	13.84	13.50	13.60	13.66	13.66	13.59	13.52	13.46
Russia	11.09	10.96	10.70	10.56	10.60	10.64	10.63	10.63	10.63
Azerbaijan	0.67	0.62	0.60	0.59	0.62	0.60	0.58	0.56	0.55
Kazakhstan	1.82	1.93	1.88	2.14	2.13	2.13	2.09	2.04	2.00
Others	0.33	0.33	0.32	0.31	0.30	0.30	0.29	0.28	0.27
Asia	6.90	6.94	6.95	6.99	6.97	6.88	6.84	6.74	6.62
China	4.18	4.27	4.34	4.39	4.43	4.39	4.34	4.26	4.20
Malaysia	0.56	0.56	0.54	0.54	0.54	0.51	0.49	0.48	0.47
India	0.72	0.70	0.70	0.71	0.69	0.68	0.68	0.67	0.67
Indonesia	0.63	0.63	0.60	0.59	0.58	0.59	0.60	0.61	0.62
Others	0.81	0.78	0.78	0.75	0.73	0.71	0.73	0.72	0.67
Europe	0.11	0.10	0.09	0.09	0.08	0.08	0.08	0.09	0.09
Americas	5.67	6.20	6.44	6.83	7.18	7.27	7.72	7.78	7.59
Brazil	3.12	3.49	3.44	3.73	3.92	3.82	4.07	3.93	3.77
Argentina	0.71	0.77	0.83	0.89	0.95	1.02	1.10	1.18	1.26
Colombia	0.76	0.79	0.79	0.76	0.73	0.71	0.68	0.65	0.63
Guyana	0.28	0.39	0.62	0.70	0.86	1.05	1.18	1.19	1.13
Others	0.80	0.77	0.77	0.75	0.71	0.68	0.69	0.82	0.80
Middle East	3.16	3.12	3.08	3.13	3.17	3.26	3.47	3.62	3.70
Oman	1.07	1.06	1.00	1.00	1.03	1.03	1.04	1.04	1.04
Qatar	1.80	1.81	1.84	1.88	1.91	2.00	2.21	2.37	2.46
Others	0.29	0.26	0.24	0.24	0.23	0.22	0.22	0.21	0.20
Africa	2.52	2.51	2.50	2.48	2.47	2.61	2.62	2.53	2.40
Angola	1.18	1.14	1.16	1.06	1.09	1.15	1.16	1.12	1.05
Egypt	0.60	0.60	0.57	0.53	0.51	0.49	0.46	0.44	0.42
Others	0.74	0.78	0.77	0.88	0.88	0.98	0.99	0.96	0.93
Total Non-OECD	32.26	32.72	32.57	33.12	33.52	33.76	34.31	34.27	33.87
Processing gains ⁵	2.32	2.36	2.39	2.40	2.46	2.48	2.47	2.46	2.47
Global biofuels	2.90	3.14	3.40	3.52	3.68	3.79	3.91	4.02	4.08
TOTAL NON-OPEC	66.93	69.36	70.26	71.61	72.43	72.83	73.41	73.25	72.65
TOTAL SUPPLY	100.05	102.32	103.05						

1 Neutral Zone production is already included in Saudi Arabia and Kuwait production, split 50/50 between the two countries.

2 Includes condensates reported by OPEC countries, oil from non-conventional sources, e.g. GTL in Nigeria, and non-oil inputs to Saudi Arabian MTBE.

3 OPEC data based on current membership throughout the time series.

4 Comprises crude oil, condensates, NGLs and oil from non-conventional sources.

5 Net volumetric gains and losses in refining and marine transportation losses.

Table 3a
SELECTED UPSTREAM PROJECT START-UPS

Country	Project	Peak Capacity (kb/d)	Start Year	Country	Project	Peak Capacity (kb/d)	Start Year
OECD Americas				OECD Europe			
US	Anchor	75	2024	Norway	Eldfisk North	30	2024
US	Whale	80	2025	Norway	Tyrving	30	2024
US	Shenandoah Phase 1	60	2025	Norway	Balder X	40	2025
US	Ballymore	75	2025	Norway	Johan Castberg	220	2025
US	Leon/Castile	60	2025	Norway	Yggdrasil	100	2027
US	Pikka Phase 1 (Alaska)	80	2026	Denmark	Tyra Redevelopment	20	2024
US	Shenandoah Phase 2	60	2027	UK	Penguins	40	2025
US	Vito Infill	30	2027	UK	Rosebank	60	2028
US	Sparta	90	2028	Middle East			
US	Willow (Alaska)	150	2029	Saudi Arabia	Dammam Phase 1	25	2025
US	Kaskida	80	2029	Saudi Arabia	Berri Expansion	250	2025
Canada	Mildred Lake Extension	140	2025	Saudi Arabia	Marjan Expansion	300	2025
Canada	West White Rose	50	2026	Saudi Arabia	Zuluf Expansion	600	2026
Mexico	Trion	100	2028	Saudi Arabia	Dammam Phase 2	50	2027
Latin America				Qatar	North Field Expansion East	240	2026
Brazil	Atlanta FDS	50	2024	Qatar	North Field Expansion South	120	2028
Brazil	Mero 3 (Mal. Duque de Caxias)	180	2024	Qatar	Bul Hanine Redevelopment	60	2027
Brazil	IPB (Maria Quitéria)	100	2024	Qatar	Al Shaheen Gallaf	90	2027
Brazil	Bacalhau	220	2025	UAE	Belbazem	45	2024
Brazil	Mero 4 (Alexandre de Gusmão)	180	2025	UAE	Upper Zakum Expansion	200	2027
Brazil	Buzios 6 (P-78)	180	2025	UAE	Lower Zakum Expansion	150	2027
Brazil	Buzios 7 (Alm. Tamandaré)	220	2025	UAE	Bab South East Expansion	130	2027
Brazil	Buzios 8 (P-79)	180	2026	UAE	Umm Shaif Expansion	130	2027
Brazil	Buzios 9 (P-80)	225	2027	Africa			
Brazil	Buzios 10 (P-82)	225	2027	Senegal	Sangomar Phase 1 (SNE)	100	2024
Brazil	Buzios 11 (P-83)	225	2027	Niger	Agadem Phase 2	50	2024
Brazil	Maromba	60	2027	Nigeria	Bonga North	110	2029
Brazil	Raia (BM-C-33)	125	2028	Cote d'Ivoire	Baleine Phase 2	30	2024
Brazil	Atapu 2 (P-84)	225	2029	Angola	Begonia	50	2025
Brazil	Gato do Mato	100	2029	Angola	Ndungu	40	2026
Brazil	Sepia 2 (P-85)	225	2030	Angola	Agogo Phase 3	120	2027
Guyana	Stabroek Phase 4 (Yellowtail)	250	2025	Angola	Kaminho	50	2028
Guyana	Stabroek Phase 5 (Uaru)	250	2026	Uganda	Lake Albert (Kingfisher and Tilenga)	200	2026
Guyana	Stabroek Phase 6 (Whiptail)	250	2028	Asia			
Suriname	Block 58	220	2028	China	Liuhua	30	2024
Eurasia				China	Lufeng	20	2024
Azerbaijan	Azeri Central East (ACE)	100	2024	China	Wushi	30	2024
Kazakhstan	Tengizchevroil FGP	260	2025	India	KG-DWN-98/2 (Cluster-2)	50	2024
Kazakhstan	Kashagan Expansion Phase 1	25	2026	Viet Nam	Lac Da Vang	30	2027
				Viet Nam	White Whale	25	2027

Table 3b
SELECTED UPSTREAM PRE-SANCTION PROJECT

(Projects with procurement/engineering started and first oil potentially by 2030)

Country	Project	Peak Capacity (kb/d)	Sanction Year	Start Year
OECD Americas				
US	Tiber	120	2025	2029
US	Mad Dog Spar	40	2025	2027
US	Ocotillo	50	2026	2027
US	Winterfell extension	40	2025	2027
US	Perdido Spar	80	2026	2028
US	Whale Phase 2	70	2026	2028
US	Pikka Phase 2	40	2027	2030
Mexico	Polok-Chinwol	50	2026	2029
Mexico	Zama	150	2026	2030
Mexico	KMZ expansion	100	2026	2028
Latin America				
Guyana	Stabroek Phase 7 (Hammerhead)	150	2025	2030
Guyana	Stabroek Phase 8 (Longtail)	200	2026	2030
Brazil	BRC/CRT Revit	100	2025	2030
Brazil	Bacalhau Phase 2	100	2026	2030
OECD Europe				
Norway	Fram (extension)	40	2025	2028
Norway	Troll (extension)	30	2025	2029
Norway	Johan Castberg Phase 2	30	2026	2028
Norway	Asgard	30	2026	2028
Norway	Balder/Ringhorne	30	2026	2028
Norway	Yggdrasil (extension)	40	2026	2029
Norway	Johan Sverdrup Phase 3	60	2025	2028
Norway	Wisting	80	2026	2029
UK	Cambo	50	2026	2029
Africa				
Cote d'Ivoire	Baleine Phase 3	100	2025	2028
Cote d'Ivoire	Calao	100	2026	2030
Ghana	Pecan	70	2026	2030
Namibia	Venus	150	2026	2030
Senegal	Sangomar Phase 2	80	2025	2029
Angola	Block 32	80	2025	2029
Asia				
Australia	Dorado	80	2027	2030
Eurasia				
Kazakhstan	Kashagan Phase 2	50	2026	2030

Table 3c
NON-OPEC SUPPLY: OIL MARKET REPORT AND WEO DEFINITIONS
(million barrels per day)

		Calculation	2022	2023	2024	2025	2026	2027	2028	2029	2030
Oil 2024 Report definitions											
NON-OPEC SUPPLY			66.9	69.4	70.3	71.6	72.4	72.8	73.4	73.2	72.7
Processing gains			2.3	2.4	2.4	2.4	2.5	2.5	2.5	2.5	2.5
Global biofuels			2.9	3.1	3.4	3.5	3.7	3.8	3.9	4.0	4.1
NON-OPEC PRODUCTION (excl. processing gains and biofuels)		1	61.7	63.9	64.5	65.7	66.3	66.6	67.0	66.8	66.1
Crude		2	50.4	51.9	52.0	52.8	53.2	53.2	53.4	53.0	52.3
of which: Condensate		3	4.3	4.5	4.6	4.5	4.5	4.6	4.7	4.8	4.9
Tight oil		4	8.7	9.6	10.2	10.6	10.6	10.8	10.9	11.0	11.2
Un-upgraded bitumen		5	2.0	2.0	2.1	2.1	2.1	2.2	2.3	2.4	2.4
NGLs		6	9.2	9.8	10.3	10.6	10.8	11.0	11.2	11.3	11.4
Syncrude (Canada)		7	1.3	1.4	1.5	1.5	1.5	1.5	1.5	1.5	1.6
CTL, GTL, kerogen oil and additives ¹		8	0.7	0.7	0.7	0.7	0.8	0.8	0.8	0.9	0.9
World Energy Outlook definitions											
NON-OPEC PRODUCTION (excl. processing gains and biofuels)		=1	61.7	63.9	64.5	65.7	66.3	66.6	67.0	66.8	66.1
Conventional			49.0	50.1	50.0	50.8	51.3	51.2	51.4	51.0	50.2
Crude oil		=2-3-4-5	35.5	35.8	35.1	35.6	35.9	35.6	35.5	34.8	33.9
Natural gas liquids (total)		=3+6	13.5	14.3	14.9	15.2	15.4	15.7	15.9	16.1	16.3
Unconventional			12.7	13.8	14.5	14.9	15.0	15.3	15.6	15.8	15.9
EHOB (incl. syncrude) ²		=5+7	3.3	3.4	3.6	3.6	3.6	3.7	3.9	3.9	3.9
Tight oil		=4	8.7	9.6	10.2	10.6	10.6	10.8	10.9	11.0	11.2
CTL, GTL, kerogen oil and additives ¹		=8	0.7	0.7	0.7	0.7	0.8	0.8	0.8	0.9	0.9

¹ CTL = coal to liquids; GTL = gas to liquids.

² Extra-heavy oil and bitumen

Table 4
WORLD REFINERY CAPACITY ADDITIONS AND CLOSURES
(thousand barrels per day)

	2024	2025	2026	2027	2028	2029	2030	Total
Refining Capacity Additions and Expansions¹								
OECD Americas	50	-222	-145	0	0	0	0	-367
OECD Europe	-120	-367	0	0	0	0	0	-367
OECD Asia Oceania	-120	0	0	0	0	0	0	0
Eurasia	70	127	50	0	0	0	0	177
Non-OECD Europe	0	0	0	0	0	0	0	0
China	310	389	273	0	-50	0	320	932
Other Asia	93	40	684	60	206	85	180	1255
Non-OECD Americas	0	15	0	0	0	115	0	130
Middle East	280	150	242	0	170	0	0	562
Africa	750	30	50	30	110	0	0	220
Total World	1313	162	1154	90	436	200	500	2542
Upgrading Capacity Additions²								
OECD Americas	0	-329	0	0	0	0	0	-329
OECD Europe	30	-36	0	0	0	0	0	-36
OECD Asia Oceania	-27	0	0	0	0	0	0	0
Eurasia	0	362	181	0	0	0	0	543
Non-OECD Europe	0	0	0	0	0	0	0	0
China	280	20	-183	0	-118	0	0	-281
Other Asia	5	232	196	217	57	40	0	742
Non-OECD Americas	0	0	0	0	85	0	0	85
Middle East	120	0	197	0	0	0	0	197
Africa	280	0	0	0	26	0	0	26
Total World	688	249	391	217	50	40	0	947
Desulphurisation Capacity Additions³								
OECD Americas	0	-557	0	0	0	0	0	-557
OECD Europe	38	-57	0	0	0	0	0	-57
OECD Asia Oceania	-109	0	0	0	0	0	0	0
Eurasia	0	110	0	0	0	0	0	110
Non-OECD Europe	0	0	0	0	0	0	0	0
China	376	162	-84	0	-42	0	0	36
Other Asia	97	198	317	153	162	0	0	830
Non-OECD Americas	0	0	-37	0	80	0	0	43
Middle East	0	0	261	0	0	0	0	261
Africa	263	0	0	0	43	0	0	43
Total World	666	-144	457	153	244	0	0	710

¹ Comprises new refinery projects or expansions to existing Crude distillation units including condensate splitter additions. Assumes zero capacity creep.

Table 4a
CHANGES FROM OIL 2024
(thousand barrels per day)

	2024	2025	2026	2027	2028	2029	2030	Total
Refining Capacity Additions and Expansions¹								
OECD Americas	170	-299	-145					-274
OECD Europe		-9						-9
OECD Asia Oceania								
Eurasia		127						127
Non-OECD Europe								
China	-140	329	-258		70		320	321
Other Asia		-220	72	-140	82	30	180	4
Non-OECD Americas					-115	115		
Middle East	110	20	10		70			210
Africa	60	-30	50	30	110			220
Total World	200	-82	-271	-110	217	145	500	599
Upgrading Capacity Additions²								
OECD Americas		-132						-132
OECD Europe								
OECD Asia Oceania	-27							-27
Eurasia		40						40
Non-OECD Europe								
China	106	20						126
Other Asia		-170		106				-64
Non-OECD Americas								
Middle East	40		55					95
Africa		-26		-29	26			-29
Total World	119	-269	55	77	26			8
Desulphurisation Capacity Additions³								
OECD Americas		-155						-155
OECD Europe								
OECD Asia Oceania	-109							-109
Eurasia		40						40
Non-OECD Europe								
China	74	162						236
Other Asia	22	-22						
Non-OECD Americas								
Middle East			-22					-22
Africa		-43		-38	43			-38
Total World	-12	-18	-22	-38	43			-47

¹ Comprises new refinery projects or expansions to existing facilities including condensate splitter additions. Assumes zero capacity creep.

Table 4b
SELECTED REFINERY CRUDE DISTILLATION CHANGES LIST

Country	Project	Capacity (kb/d)	Year	Country	Project	Capacity (kb/d)	Year
OECD Americas				Asia			
Mexico	Dos Bocas	170	2025	China	Huajin Petchem (Panjin II)	320	2026
United States	Wilmington	-140	2025	China	Fujian Gulei Petchem (Gulei II)	320	2030
United States	Benicia	-150	2026	China	Zhenhai Refining and Chemical	240	2025
United States	Houston Lyondell	-260	2025	China	Yulong Petrochemical	210	2025
OECD Europe				China	Daxie Petrochemical	130	2025
Germany	Gelsenkirchen	-80	2025	China	Qilu Petchem	70	2028
Germany	Wesseling	-150	2025	China	Dalian (II) WEPEC	-200	2025
United Kingdom	Grangemouth	-140	2025	India	Panipat	200	2026
Middle East				India	Nagapattinam	180	2030
Bahrain	Sitra	120	2026	India	Barmer	180	2026
Iran	Persian Gulf Star (Bandar Abbas IV)	120	2025	India	Barauni, Bihar	130	2027
Iraq	Dhi Qar	100	2028	India	Numaligarh, Assam	120	2026
Iraq	Diwaniya	70	2028	India	Koyali, Gujarat	80	2026
Iraq	Missan	70	2026	India	Barauni, Bihar	-70	2027
NON-OECD Americas				Indonesia	Balikpapan (Ref Unit V)	100	2026
Brazil	RNEST	120	2029	Thailand	Sriracha I	100	2028
Africa							
Algeria	Hassi-Messaoud	110	2028				

Note: Only includes refinery capacity changes (additions or closures) above 70 kb/d. The data is rounded to the nearest 10 kb/d.

Table 5
WORLD ETHANOL PRODUCTION¹

(thousand barrels per day)

	2022	2023	2024	2025	2026	2027	2028	2029	2030
OECD North America	1031	1049	1086	1074	1068	1076	1079	1089	1083
United States	1002	1019	1055	1038	1029	1037	1040	1048	1042
Canada	29	30	31	36	38	39	40	41	41
OECD Europe	112	114	123	130	140	149	164	183	193
Austria	5	5	4	4	4	4	4	4	4
Belgium	12	12	12	12	12	12	12	12	12
France	20	21	22	23	24	26	28	30	30
Germany	13	12	14	14	15	16	16	16	16
Italy	0	1	2	4	4	4	4	4	4
Netherlands	10	10	11	12	16	20	23	38	46
Poland	7	7	8	9	10	11	12	14	15
Spain	9	10	10	10	14	15	15	15	15
UK	8	8	9	9	9	10	19	19	19
OECD Pacific	4	4	4	4	6	9	10	13	13
Australia	4	4	4	4	4	4	4	6	6
Total OECD	1147	1167	1213	1209	1214	1235	1254	1284	1288
Eurasia	0	0	0	0	0	0	0	0	0
Non-OECD Europe	2	2	1	1	2	2	2	2	2
China	66	68	86	87	87	87	87	88	93
Middle East	0	0	0	0	0	0	0	0	0
Africa	5	5	5	5	5	5	5	3	3
Other Asia	112	127	156	181	202	219	243	259	278
India	79	93	117	136	154	166	185	199	212
Indonesia	0	0	1	2	3	5	7	9	10
Malaysia	0	0	0	0	0	0	0	0	0
Philippines	7	7	7	7	7	7	7	7	7
Singapore	1	2	5	7	7	9	11	11	14
Thailand	25	25	26	29	30	32	32	33	34
Latin America	565	646	678	693	721	752	774	810	820
Argentina	20	20	18	21	21	22	22	23	23
Brazil	528	607	640	652	680	710	732	768	777
Colombia	6	7	7	7	7	7	7	7	7
Total Non-OECD	751	849	927	967	1017	1065	1111	1163	1195
Total World	1898	2016	2140	2176	2231	2300	2364	2447	2483

¹ Volumetric production; to convert to energy adjusted production, ethanol is assumed to have 2/3 energy content of conventional gasoline.

Table 5a
WORLD BIODIESEL PRODUCTION¹
 (thousand barrels per day)

	2022	2023	2024	2025	2026	2027	2028	2029	2030
OECD North America	209	288	335	340	390	378	389	399	399
United States	203	280	316	312	357	339	346	356	356
Canada	6	8	19	28	34	38	43	43	43
OECD Europe	282	293	302	313	314	330	336	342	344
Austria	6	10	11	11	11	11	11	11	11
Belgium	3	4	4	3	3	2	1	1	2
France	27	26	35	38	39	39	39	39	39
Germany	66	74	72	72	64	64	64	64	64
Italy	23	25	25	25	25	25	25	25	25
Netherlands	38	40	39	38	44	50	53	54	56
Poland	19	19	21	21	21	22	21	23	23
Spain	35	29	32	37	40	41	43	45	45
UK	13	13	13	13	13	13	13	13	13
OECD Pacific	15	16	15	15	15	15	14	14	14
Australia	0	0	0	0	0	0	0	0	0
Total OECD	506	597	653	667	719	723	739	756	757
Eurasia	0	0	0	0	0	0	0	0	0
Non-OECD Europe	15	17	13	13	14	14	14	14	14
China	42	43	43	37	37	40	43	36	36
Middle East	2	2	2	2	2	2	2	2	2
Africa	2	2	1	1	1	1	2	2	2
Other Asia	270	303	351	393	421	451	470	482	497
India	3	3	3	3	3	3	3	4	4
Indonesia	187	212	257	299	311	330	342	350	360
Malaysia	24	28	31	34	43	44	44	45	46
Philippines	4	4	4	5	6	7	7	7	8
Singapore	29	33	32	28	28	33	32	33	35
Thailand	23	23	24	24	29	35	40	42	44
Latin America	163	165	196	229	250	260	280	286	291
Argentina	37	19	24	29	29	29	30	30	30
Brazil	108	130	156	176	201	205	222	228	233
Colombia	13	13	13	13	13	13	14	14	14
Total Non-OECD	492	532	607	677	727	769	811	822	842
Total World	999	1128	1260	1344	1445	1492	1550	1578	1599

¹ Biodiesel includes renewable diesel.

Table 6
OPEC+ Crude Oil and NGL¹ Capacities² to 2030

(million barrels per day)

	2024	2030	2024	2030
	Crude Capacity	Crude Capacity	NGL Capacity	NGL Capacity
Algeria	0.99	0.91	0.45	0.42
Congo	0.27	0.23	0.01	0.01
Equatorial Guinea	0.06	0.05	0.04	0.02
Gabon	0.23	0.22	0.00	0.00
Kuwait	2.91	3.06	0.32	0.34
Nigeria	1.56	1.44	0.28	0.27
Saudi Arabia	12.1	12.3	1.76	2.73
UAE	4.30	5.02	0.95	1.21
Iraq	4.91	5.47	0.18	0.24
Iran	3.80	3.80	1.31	1.39
Libya	1.23	1.24	0.07	0.07
Venezuela	1.05	1.05	0.08	0.08
Total OPEC	33.38	34.77	5.45	6.77
Russia	9.53	9.47	1.40	1.40
Kazakhstan	1.83	1.94	0.33	0.36
Mexico	1.55	1.06	0.42	0.22
Azerbaijan	0.48	0.42	0.12	0.13
Oman	0.76	0.78	0.24	0.26
Other Non-OPEC OPEC+ ³	0.72	0.63	0.20	0.16
Total Non-OPEC	14.88	14.31	2.70	2.53
Total OPEC+	48.26	49.07	8.15	9.30

1 NGL capacities include condensates.

2 Capacity levels that can be reached within 90 days and sustained for an extended period.

3 Other Non-OPEC OPEC+ members includes Bahrain, Brunei, Malaysia, Sudan and South Sudan.

Annexes

Abbreviations and acronyms

ASEAN	Association of Southeast Asian Nations
ACE	Azeri Central East
BART	Bay Area Rapid Transit
Capex	capital expenditure
C5+	pentanes plus
CARBOB	California Reformulated Gasoline Blendstock for Oxygenate Blending
CDDs	cooling degree days
CIF	cost, insurance & freight
CDU	crude distillation unit (refinery)
CNOOC	China National Offshore Oil Company
CNPC	China National Petroleum Company
CNPE	National Energy Policy Council (Brazil)
CO ₂	carbon dioxide
CTC	crude oil to chemicals (refinery)
CTL	coal-to-liquids
DUC	drilled but uncompleted well
EIA	US Energy Information Administration
E&P	exploration and production
EOR	enhanced oil recovery
ESG	environmental, social and governance
EU	European Union
EUAs	EU Allowances
EU-ETS	EU Emissions Trading System
EVs	electric vehicles
FAME	fatty acid methyl ester
FCC	fluid catalytic cracking unit (refinery)
FGP	future growth project
FPSO	floating production, storage and offloading
FID	final investment decision
FOB	free on board
Fracking	hydraulic fracturing
GDP	gross domestic product
GFC	Global Financial Crisis (2008)
GHG	greenhouse gas
GL	US Department of the Treasury's Office of Foreign Assets Control general license

GOR	gas-oil ratio
GTL	gas-to-liquids
HCK	hydrocracker (refinery unit)
HEV	hybrid electric vehicle
HDT	hydrotreating unit
HSFO	high sulphur fuel oil
HSR	high-speed rail
HVO	hydrotreated vegetable oil
IATA	International Air Transport Association
IMO	International Maritime Organization
ICE	internal combustion engine
IEA	International Energy Agency
IOC	international oil company
IP90	Initial 90-day average oil production
IRA	Inflation Reduction Act (United States)
ITP	Iraq-Türkiye Pipeline
ITT	Ishpingo-Tambococha-Tiputini
JOA	joint operating agreement
KMZ	Ku-Maloob-Zaap
KOC	Kuwait Oil Company
KRG	Kurdistan Regional Government
LCFS	low-carbon fuel standard (used in California)
LDV	light-duty vehicle
LNG	liquefied natural gas
LPG	liquefied petroleum gas
LTO	light tight oil
MER	market exchange rates
MERAM	maximising ethane recovery and monetisation project
MGO	marine gasoil
MPG	miles per gallon
MTBE	methyl tert-butyl ether
n-butane	normal butane
NBS	China's National Bureau of Statistics
NGLs	natural gas liquids
NOC	national oil companies
NWE	Northwest Europe
OECD	Organisation for Economic Co-operation and Development
OFAC	US Department of the Treasury's Office of Foreign Assets Control
OPEC	Organization of the Petroleum Exporting Countries
OPEC-12	Algeria, Congo, Equatorial Guinea, Gabon, Iran, Iraq, Kuwait, Libya, Nigeria, Saudi Arabia, United Arab Emirates and Venezuela
OPEC+	Comprises 22 countries and includes the OPEC-12 member plus Azerbaijan, Bahrain, Brunei, Kazakhstan, Malaysia, Mexico, Oman, Russia, South Sudan, and Sudan

Opex	operating expenses
PDH	propane dehydrogenation
PDVSA	Petróleos de Venezuela, S.A.
PE	polyethylene
PP	polypropylene
PPP	purchasing power parity
PSC	production sharing contract
PVC	polyvinyl chloride
RIGI	Régimen de Incentivo para Grandes Inversiones
RPK	revenue passenger kilometre
RNEST	Brazil Abreu e Lima Refinery
RPK	rail passenger kilometre
RRR	reserve replacement ratios
SAF	sustainable aviation fuel
SECA	Sulphur Emission Control Area
STEO	short term energy outlook (US EIA)
SUVs	sports utility vehicles
TAR	turnarounds (for major maintenance)
TfL	Transport for London
TMX	Trans Mountain Expansion Project
UAE	United Arab Emirates
UCO	used cooking oil
UN	United Nations
UNCTAD	United Nations Conference on Trade and Development
USD	US Dollar
USGC	United States Gulf Coast
US PADD	US Petroleum Administration for Defence District
US PADD 1	East Coast
US PADD 2	Midwest
US PADD 3	Gulf Coast
US PADD 4	Rocky Mountain
US PADD 5	West Coast
VGO	vacuum gasoil
VLSFO	very low sulphur fuel oil
VMT	vehicle miles travelled
WCS	Western Canadian Select crude
WCSB	Western Canada Sedimentary Basin
WFH	working-from-home
WTI	West Texas Intermediate
WTO	World Trade Organization

Units of measure

bbl	barrel (of oil)
b/d	barrels per day
boe	barrels of oil equivalent
GW	gigawatt
kb/d	thousand barrels per day
km	kilometre
mb/d	million barrels per day
mb	million barrels
MMscf/d	million standard cubic feet per day
Mt	million tonnes
Mt/yr	million tonnes per year
Mtpa	million tonnes per annum
tkm	tonne-kilometres
t	tonnes

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